



MAWLANA BHASHANI
Science and Technology University,
Santosh, Tangail-1902

Engineering Academy
BPDB, BWDB, Kaptai

BPDB, BWDB, Kaptai - Seminar on Cyber security Essentials: Threats, Laws & Safe Practices in Power Sector

Cyber Security Considerations Specific to Power Grid and SCADA/OT systems

Ziaur Rahman, PhD (rahmanziaur.github.io)



LMS Course Link: <https://canvaslite.instructure.com/enroll/WT49FB>
Course Page: <https://rahmanziaur.github.io/pdb/>
GitHub Repository: <https://github.com/rahmanziaur/pdb>





“আমি খেটে খাওয়া
মানুষের কথা বলি।

মজলুম জননেতা
মওলানা আবদুল হামিদ খান ভাসানী

(১২ ডিসেম্বর ১৮৮০ - ১৭ নভেম্বর ১৯৭৬)



Acknowledgment





Mawlana Bhashani Science and Technology University



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Professor



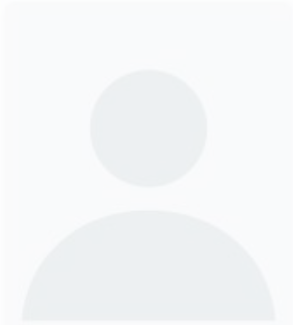
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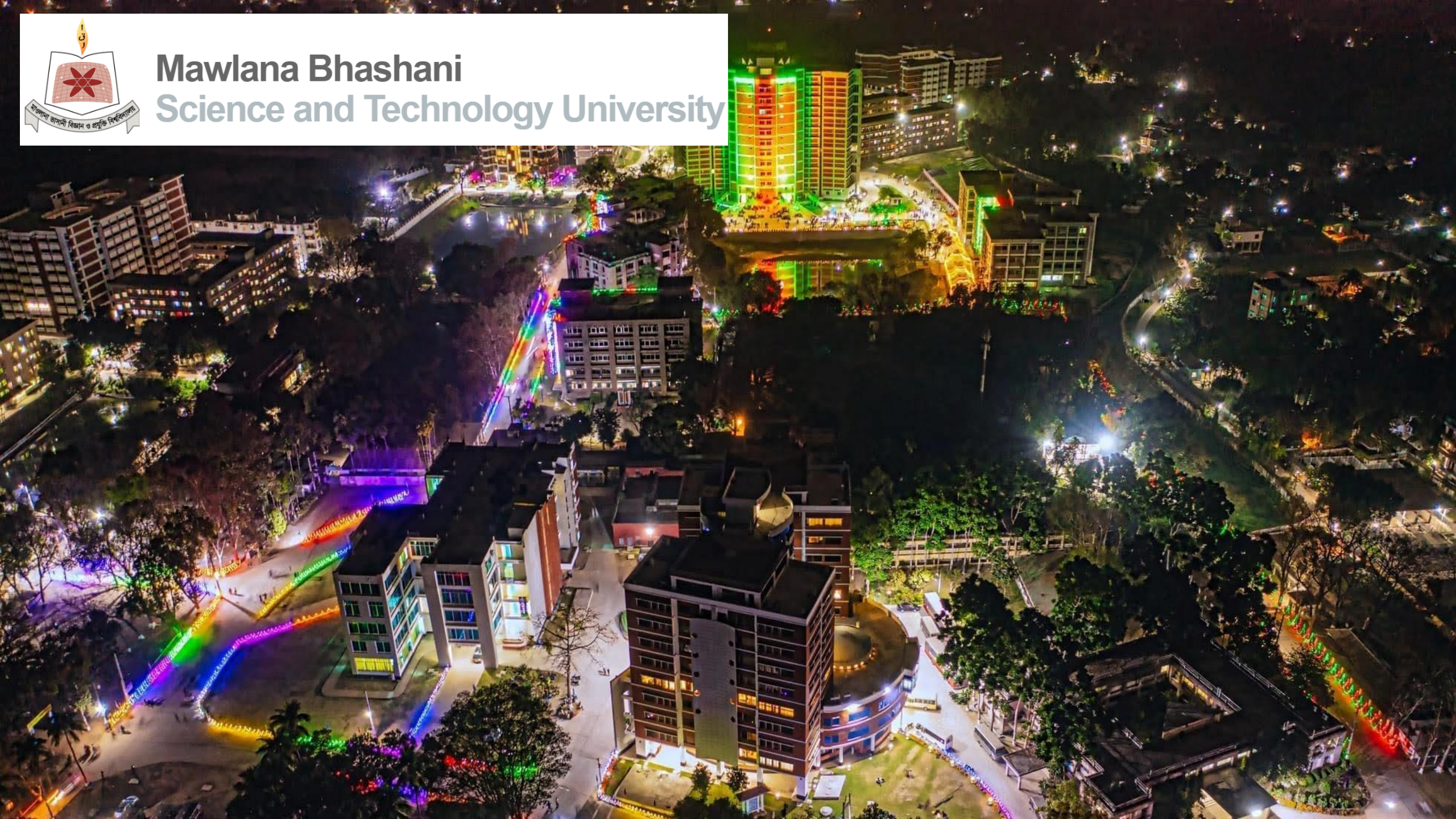
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About Me

Prof. Dr. Ziaur Rahman is an academic, researcher, and software development professional with over 12 years of experience in higher education, research, and industry-oriented technology development. He currently serves as a **Full Professor** at the Department of Information and Communication Technology (ICT), **Mawlana Bhashani Science and Technology University (MBSTU)**, Bangladesh. In addition to his academic responsibilities, he serves as the **Director of the Cyber Security Center**, the **Secretary and Primary Research Contact of the Quality Review and Ranking Committee (QRRC)**, and the **Student Advisor** of the Department of ICT at MBSTU.

As a course coordinator, Prof. Rahman designs and delivers courses in computer programming, software engineering, and cybersecurity. His research interests encompass privacy-preserving IoT security, blockchain technology, edge machine learning, and software security, with a strong focus on developing secure, scalable, and efficient solutions for emerging Industry 4.0 challenges.

Prof. Rahman earned his **PhD in Cybersecurity** from RMIT University, Australia, where he received the prestigious **Higher Degree by Research Award** and his doctoral thesis was recognized as outstanding, placing it among the **top 5% of submissions**. Following his doctoral studies, he served as a **Research Fellow in Cybersecurity** at Queensland University of Technology (QUT) and Charles Sturt University (CSU), Australia with the Cyber Security Cooperative Research Centre.

Passionate about bridging academia and industry, Prof. Rahman is an enthusiastic lifelong learner who has completed several professional certifications and advanced training programs, including **Infosec Certified Cloud Security Professional (CCSP)**, IBM Penetration Testing, Threat Hunting, Cryptography and Security, AWS Compliance, and Governance for AI Solutions.





 **Rahman Ziaur MACS** (He/Him) • You
ISC2® CISSP Cand | PhD in Blockchain & IoT...
1yr • Edited • 

Received the Higher Degree by Research (**#HDR**) Prize from RMIT University.
Great to share that the **RMIT STEM College** of **RMIT University** has awarded me the HDR Computing Technologies Award for my research performance and community contribution.

Professor **Karin**, Executive Dean, **RMIT School of Computing Technologies**, handed over the prize and certificate on July 6 at the Village Roadshow Theatre of **State Library Victoria**, Melbourne.

Distinguished Professor Adrian Mouritz, Dean, **RMIT School of Engineering** and Professor **Angela Carbone**, Associate DVC Learning Teaching & ...more

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Celebrate



Comment



Repost

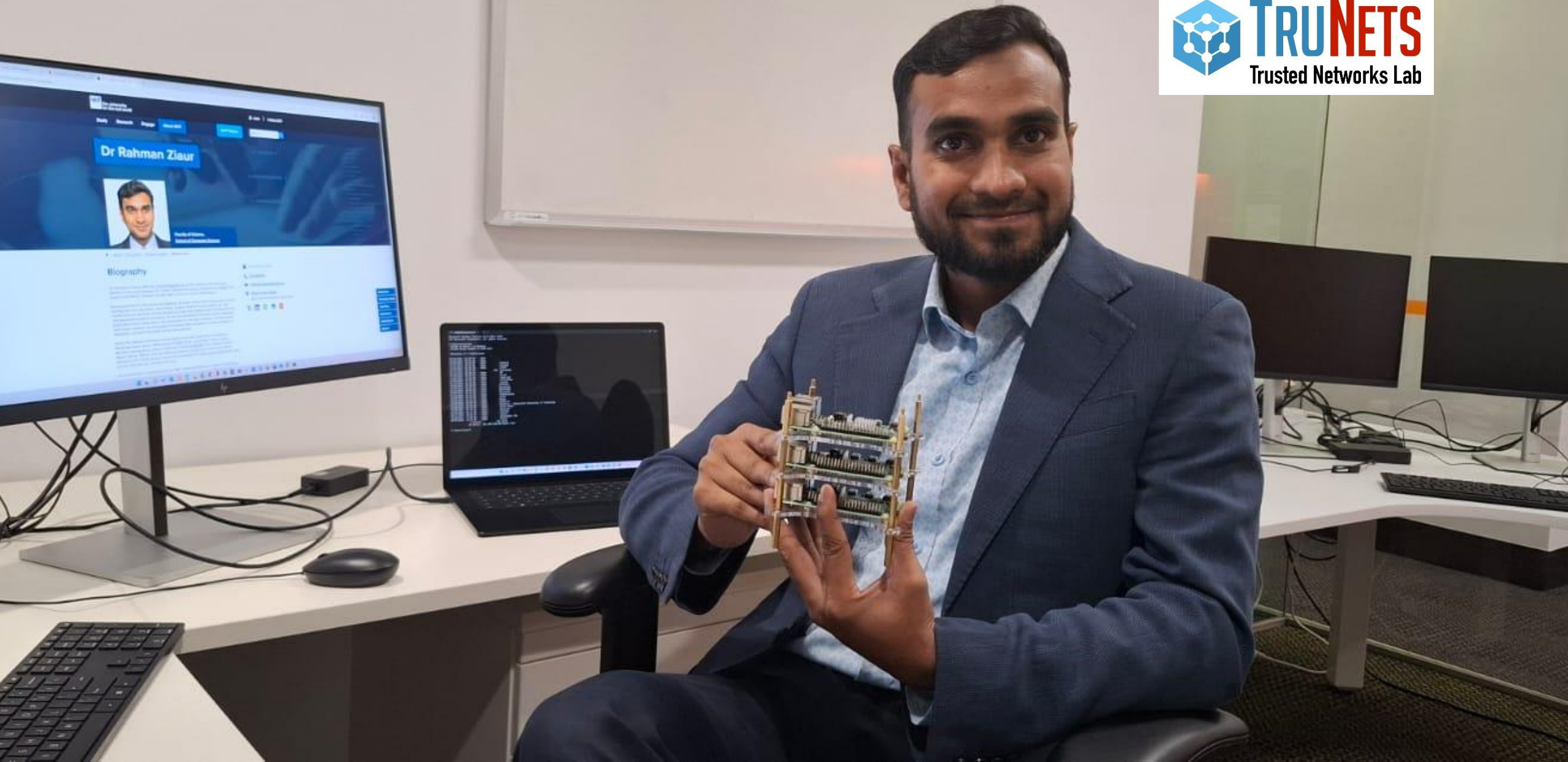


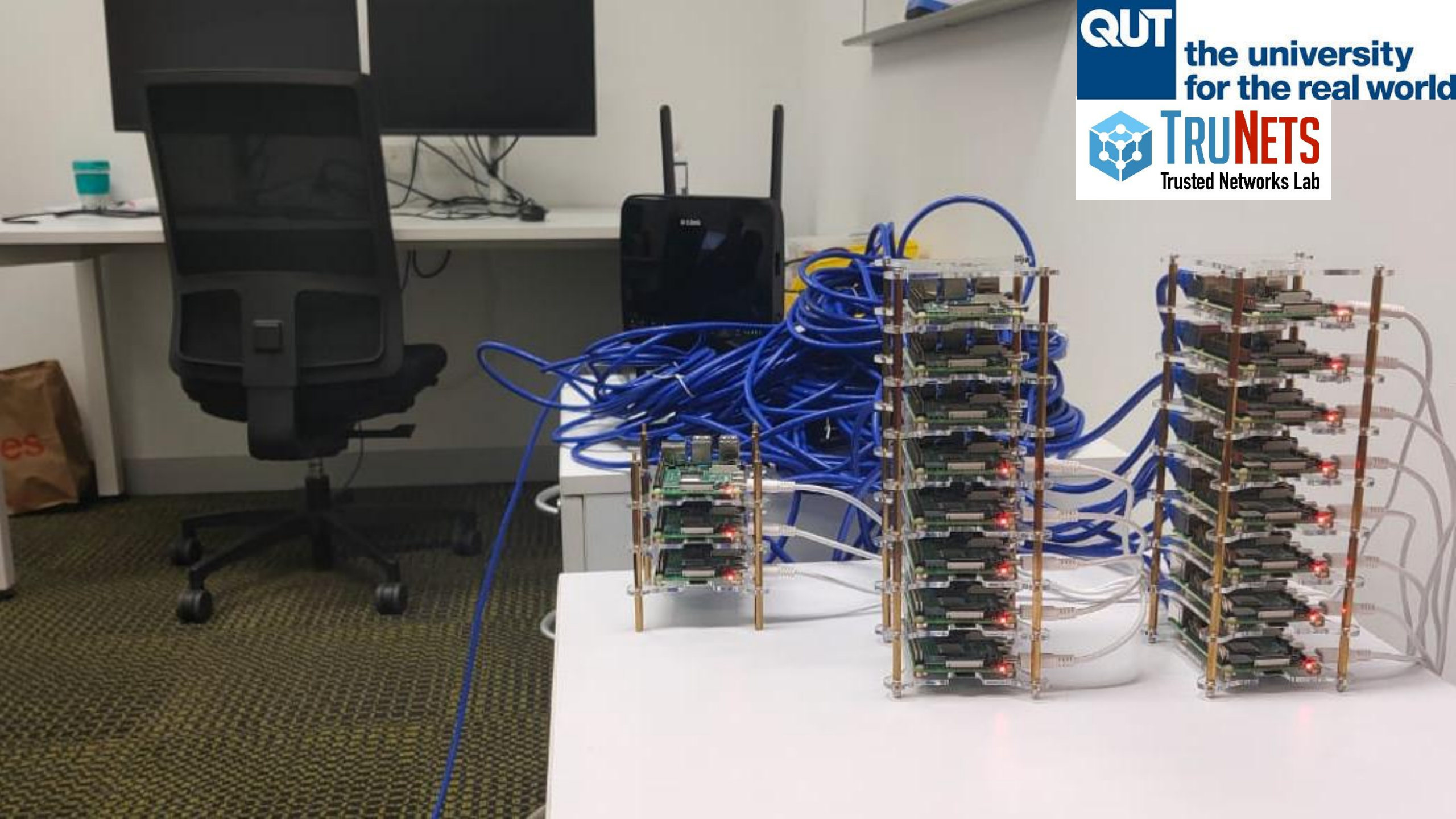
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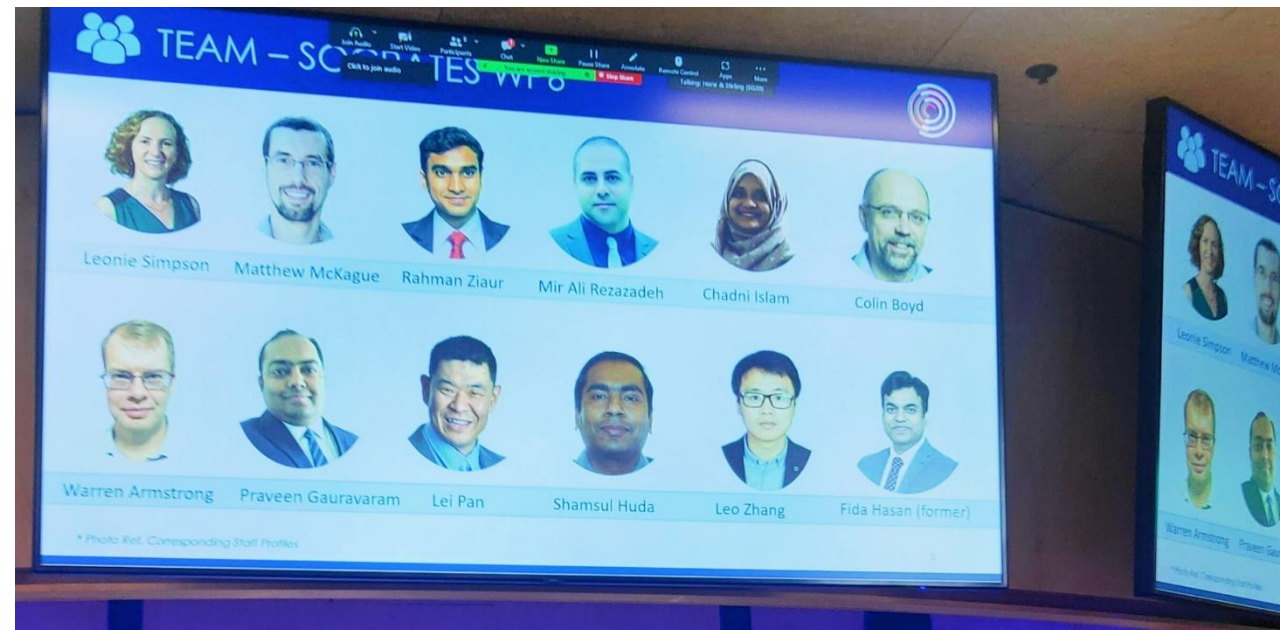
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CSCRC Awayday Showcasing 2023, Uni Adelaide



OUTLINE



- OT/IT and Automation
- SCADA, ICS, DCS
- Attacks on distribution SCADA, substations, AMI
- Prevention through role-based access
- Multi-Factor Authentication
- Meter Encryption (DLMS/COSEM)
- RTU/IED hardening and secure procurement Mitigation
- Case studies: Ukraine attacks (2015, 2016)



OUTLINE

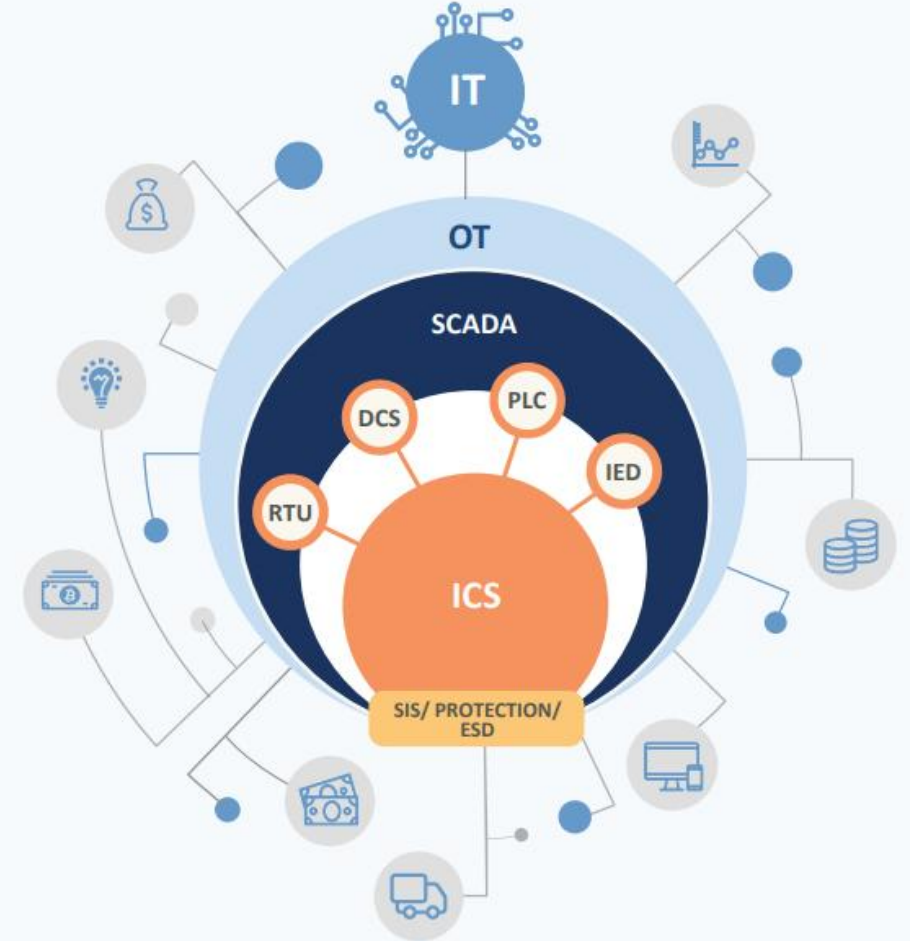


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Operational Technology (OT)

Operational Technology (OT) is hardware and software that detects or causes a change through the direct monitoring and/or control of physical devices such as valves, pumps, temperature sensors, gas sensors, etc. within industrial processes.

Industrial Control Systems (ICS) are an element of OT that are used to control valves, engines, conveyors, smelters, mixers, and other machines to regulate various process values, such as temperature, pressure, flow, chemical mixtures, and are also used to monitor them to prevent hazardous conditions.

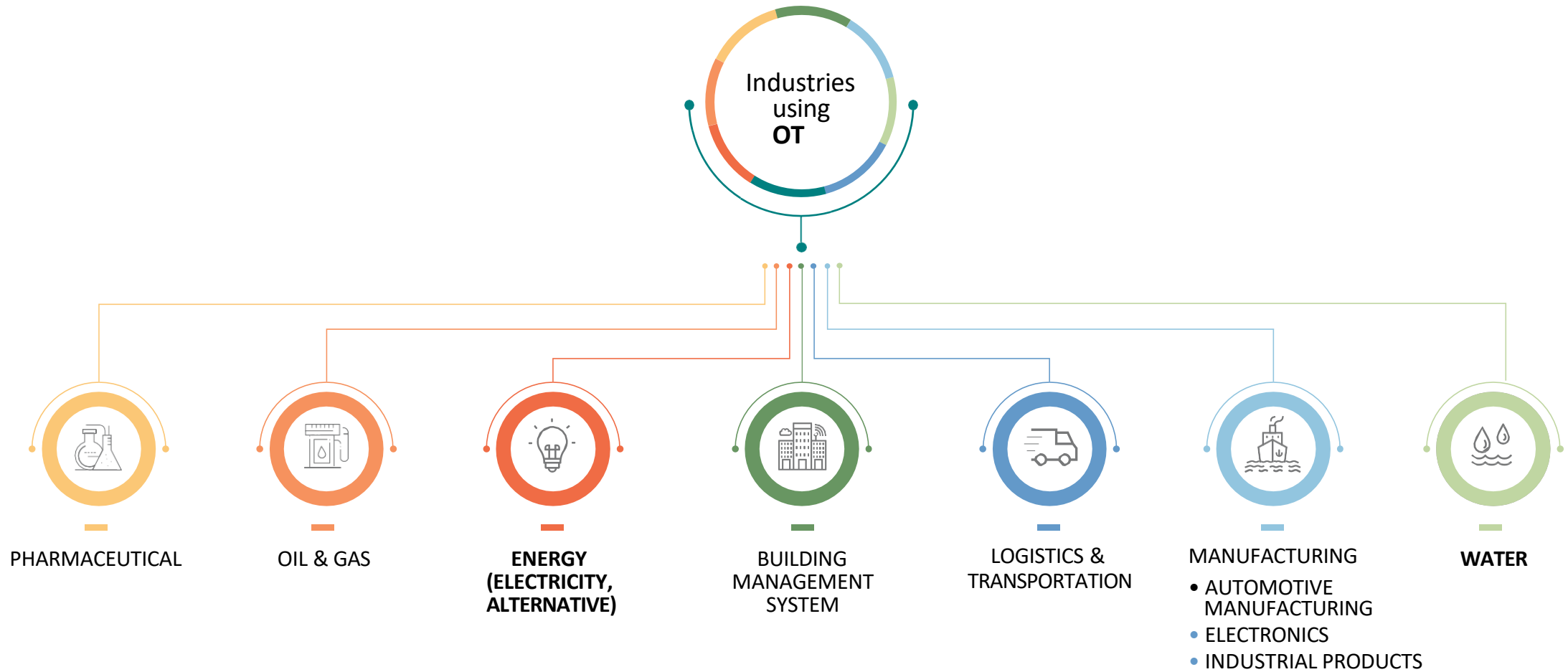


Operational Technology (OT)

Any equipment interacting with the real world, as well as the systems that control such equipment

- **Power & energy:** SCADA, PLCs, RTUs, substations, smart meters
- **Manufacturing:** PLCs, industrial robots, CNC machines
- **Water & wastewater:** pumps, valves, treatment-control systems
- **Transportation:** railway signaling and traffic-control systems
- **Oil & gas:** pipeline control and refinery systems
- **Buildings:** HVAC, access control, building-management systems

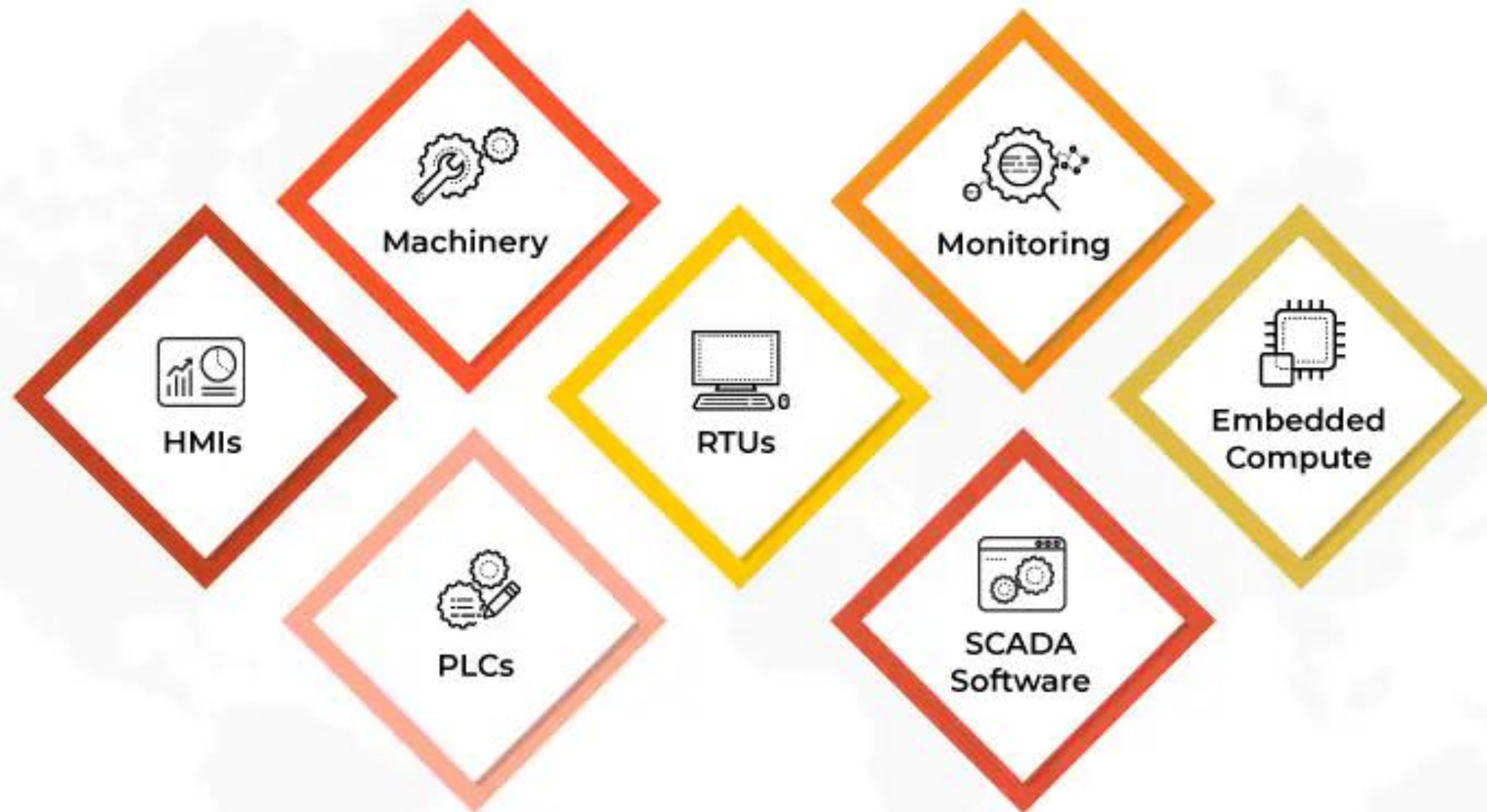
OT Security Standards Taxonomy



Information Technology



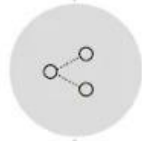
Operational Technology



IT vs. OT systems



Computers



Networks



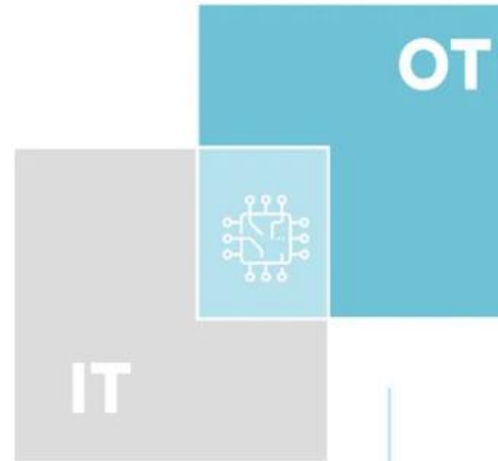
Servers



Databases



Cloud-based infrastructure



Manufacturing equipment



Transportation equipment



Energy equipment



Utilities



Controllers



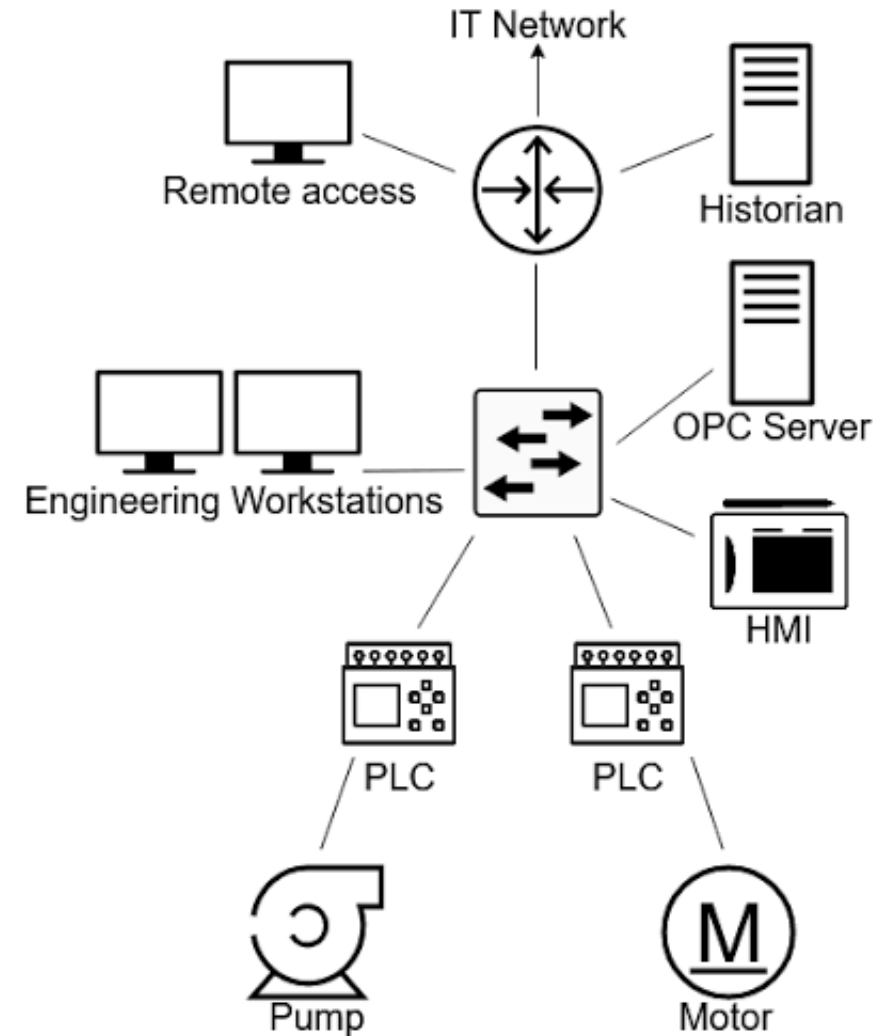
Machinery

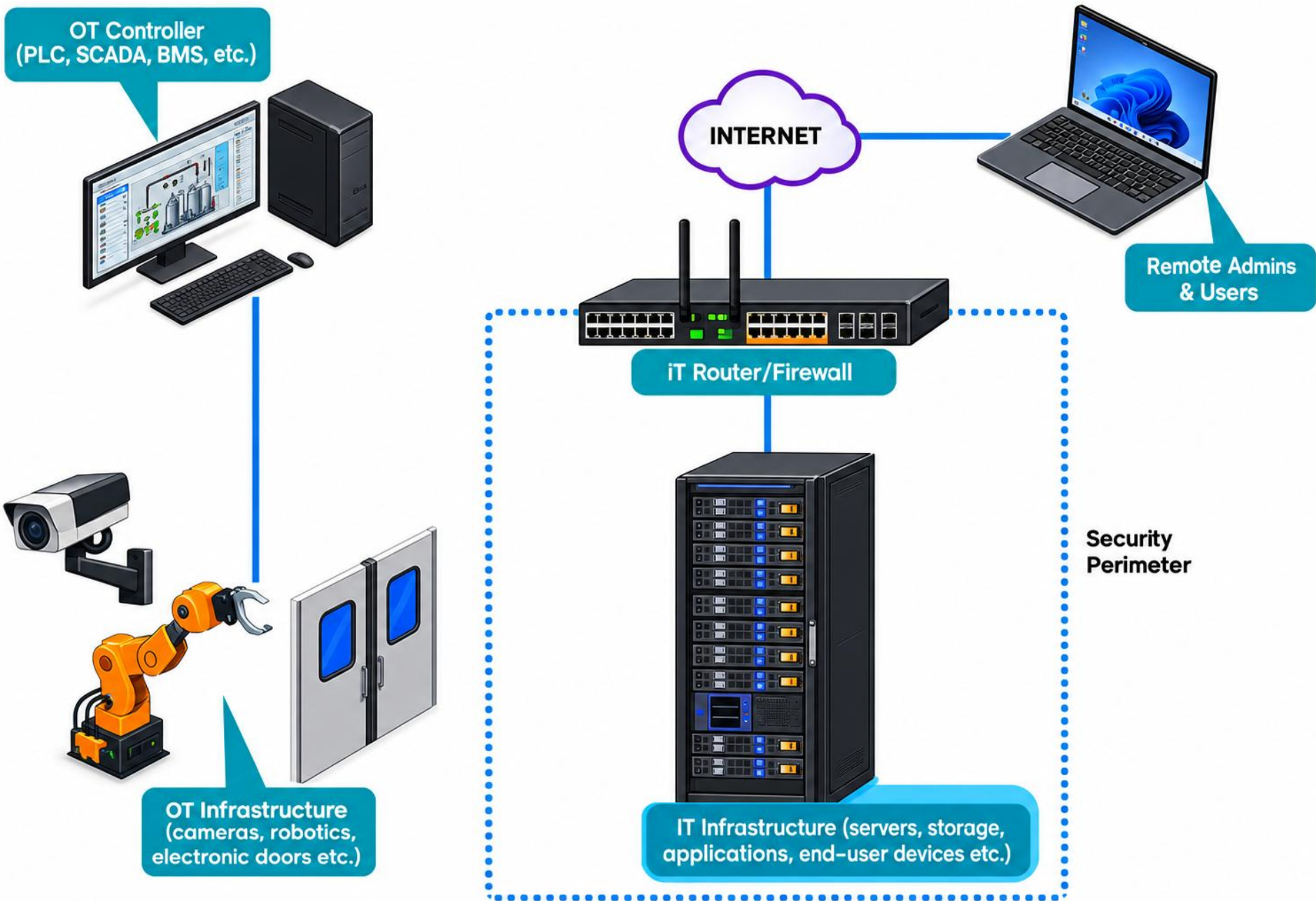


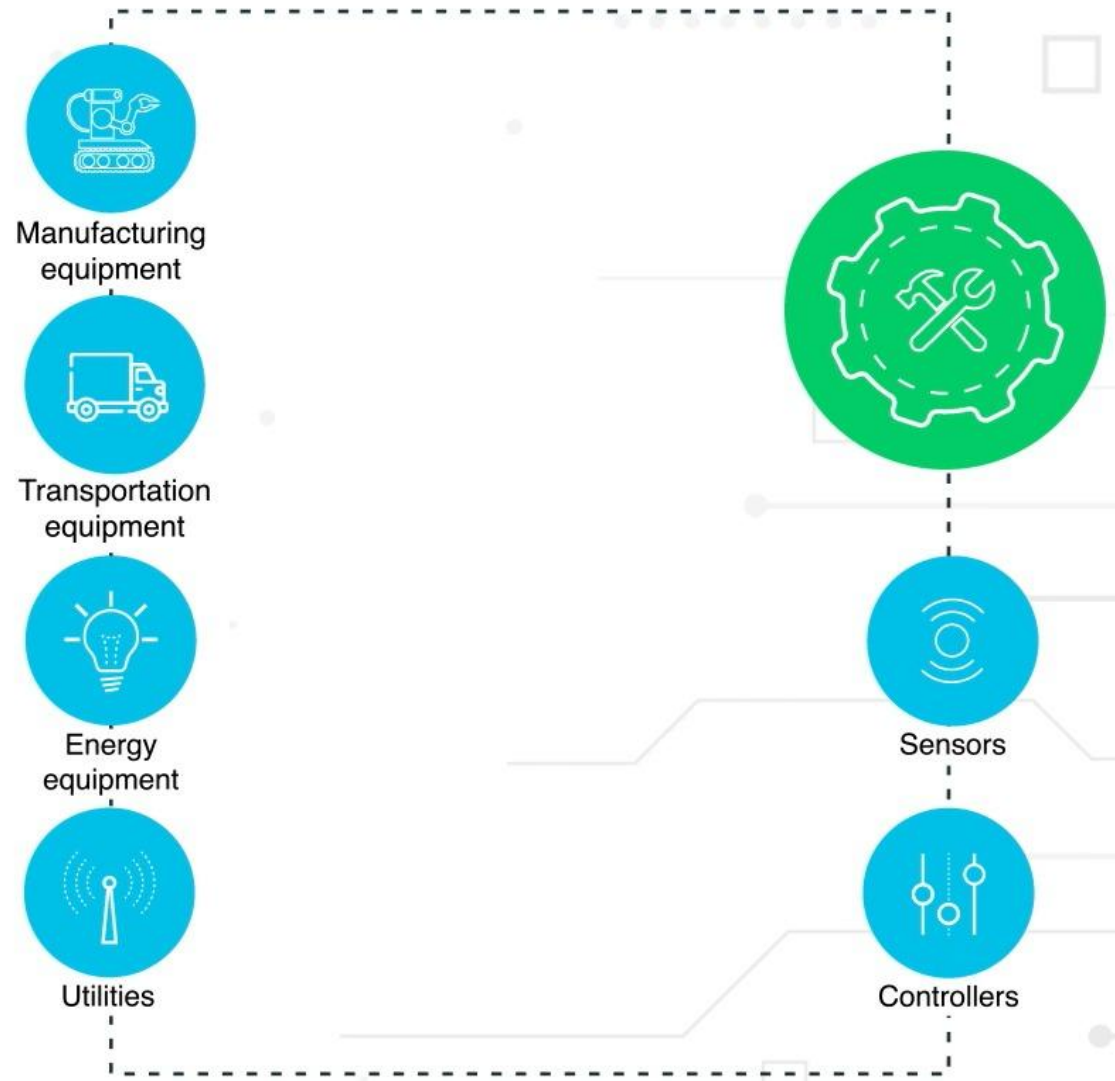
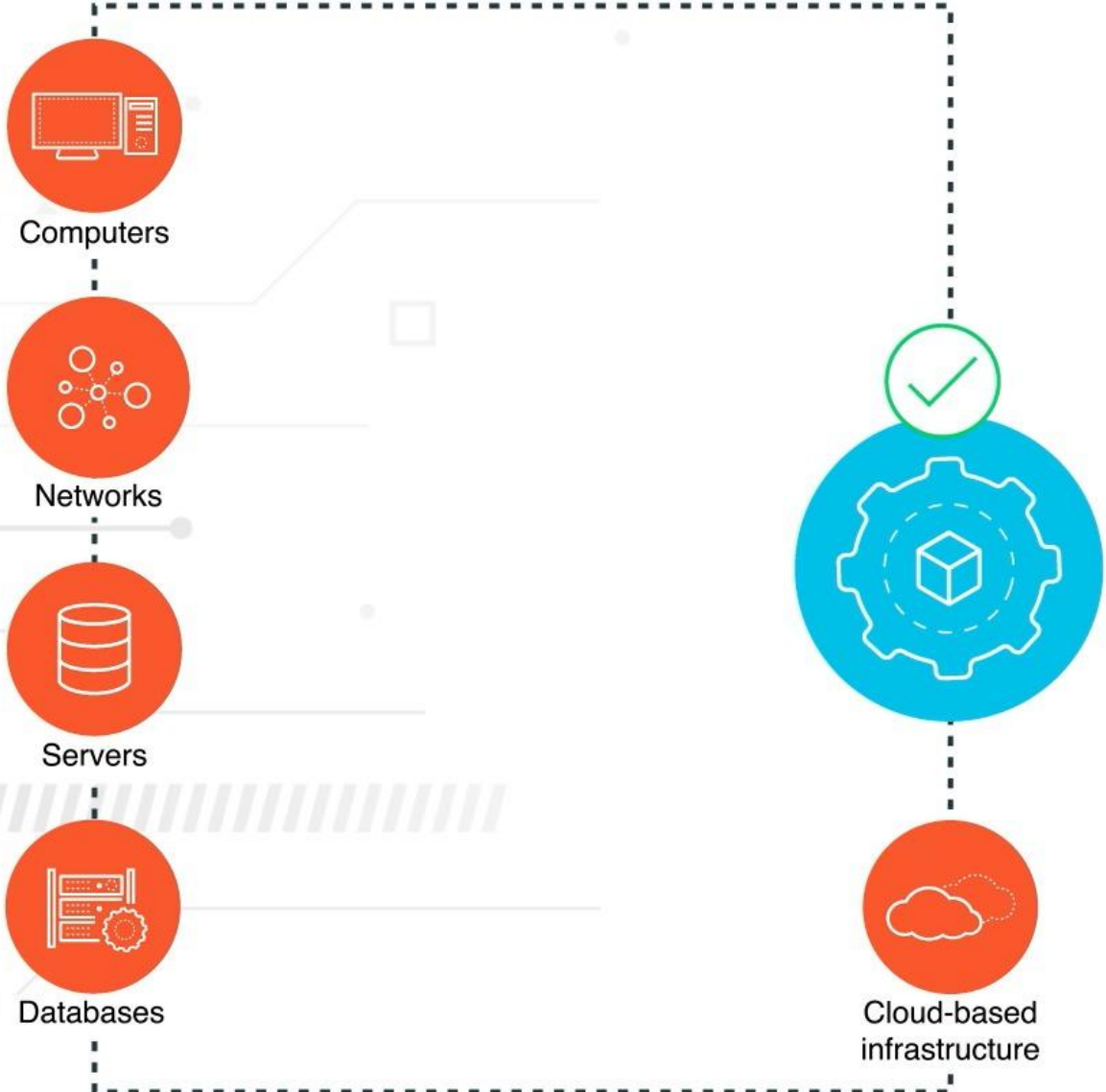
Sensors

OT Architecture

Sensors/Actuators → PLC/RTU →
SCADA/HMI → Control Network →
Operations Center







IT security priorities



1.
Confidentiality
of information



2.
Integrity of
data



3.
Availability of
data & systems

OT security priorities



1.
Availability of
operations & data



2.
Integrity of
configuration data



3.
Confidentiality

IT and OT Systems Comparison:

Connectivity and Security Requirements

	IT	OT
Connectivity Mechanisms	Via Telco, Wi-fi	Via Telco, Radio, Satellite, Powerline Carrier, Wi-Fi
Security Priority	Data security with high confidentiality	Operational uptime with high availability, safety, and integrity
Security Standards	ISO-17799, 27001, NIST SP 800-53	ISA99, NERC CIP 002-009, NIST SP 800-53, NIST SP 800-82
Security Patching	Frequent	Slow to impossible
Cyber Forensics	Available	Limited, if any
Overall Impact from Security Breaches	Business impacts	Business impacts, process fluctuations, equipment damage, environmental release, personnel safety



GDPR



HIPAA



PCI-DSS



SOX



ISO/IEC 27001



NERC CIP



IEC 62443



NIST SP 800-82



ISA/IEC 61511



CFATS

IT vs. OT

security regulatory and compliance



GDPR



HIPAA



PCI-DSS



SOX



ISO/IEC 27001



NERC CIP



IEC 62443



NIST SP 800-82



ISA/IEC 61511



CFATS



IT vs OT

IT (Information Technology)	OT (Operational Technology)
Computers and laptops	Industrial machines
Email systems	Power-grid equipment
Databases	Transformers and circuit breakers
Web applications	PLCs and RTUs
Cloud servers	SCADA systems
Office networks	Industrial control networks



Protecting critical systems
& data



Continuous monitoring



Remote management
& control



Use of firewalls, IDS,
& encryption



Similarities between IT and OT security

**IT
security**



Protect critical
systems & data



Continuous
monitoring



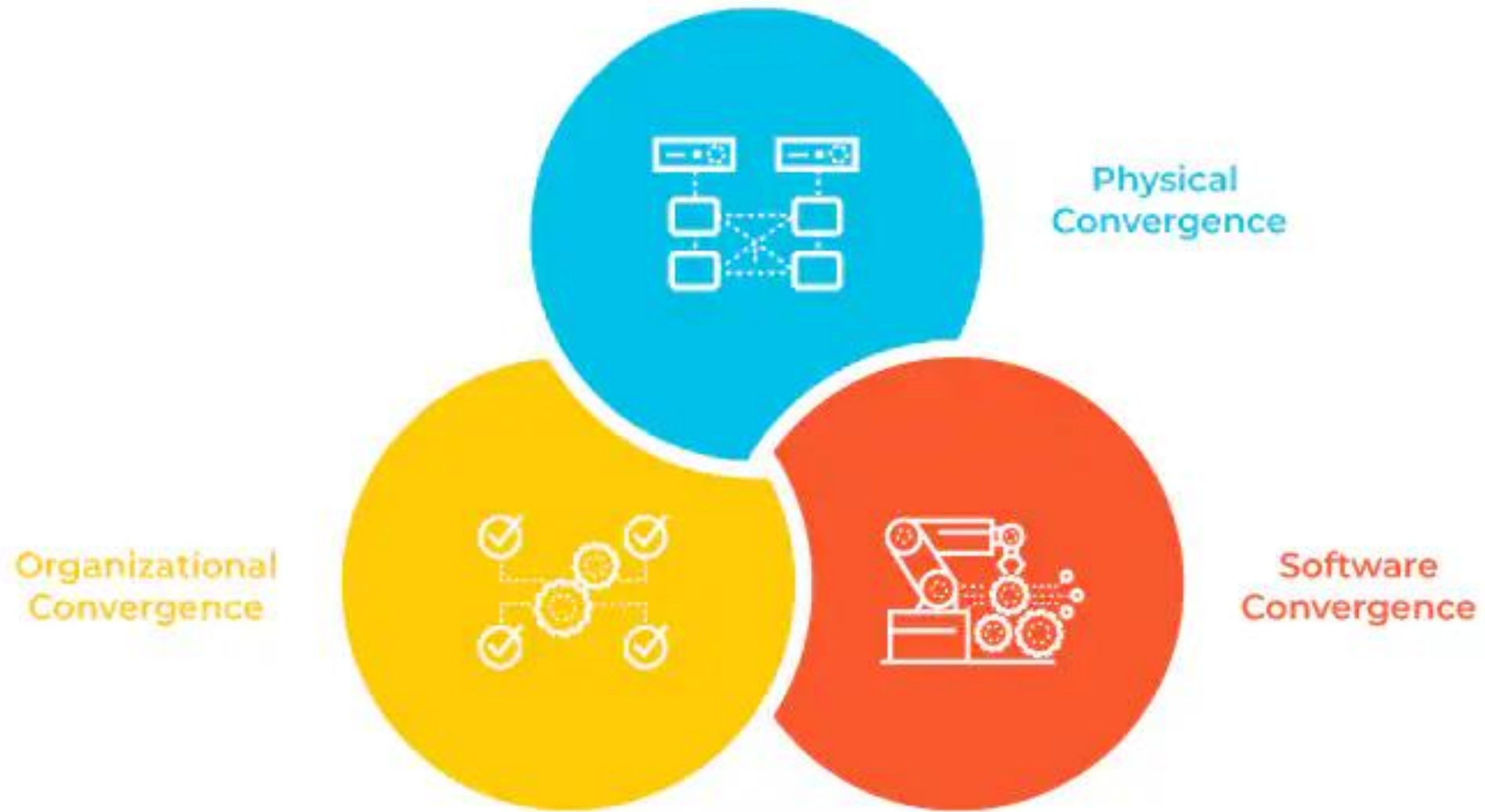
Remote management
& control



Use of firewalls, IDS,
& encryption

**OT
security**

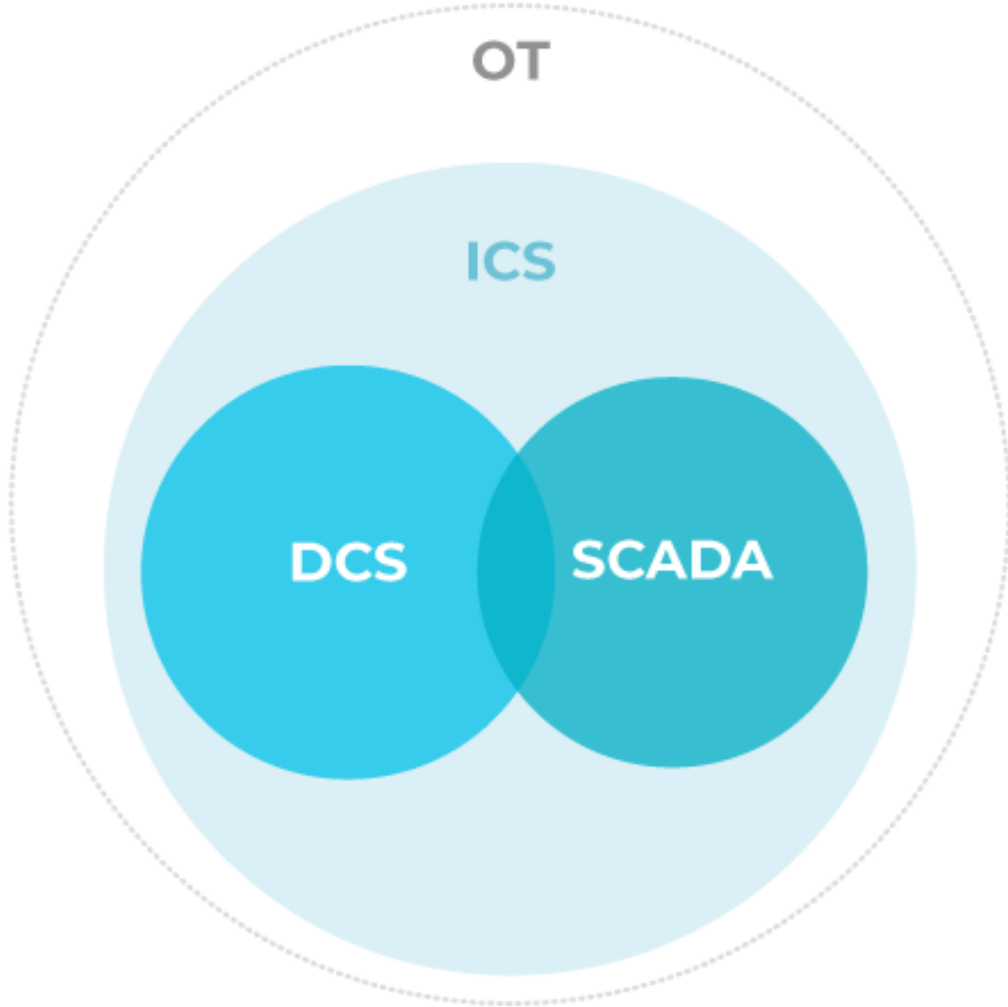
Types of IT/OT Convergence



IT vs OT Shared Concern

- Both fields aim to protect critical systems and data from unauthorized access, cyberattacks, and operational disruptions.
- growing need for continuous monitoring.
- IT and OT security now both rely heavily on remote management and control.
- IT and OT security use common technologies like firewalls, intrusion detection systems, and encryption to secure their respective environments.

OT, ICS, and SCADA Security



- Strong security measures are critical to defending **against a range of cyber threats and vulnerabilities**.
- Unlike IT systems, OT systems **prioritize operational efficiency** and safety above all.
- OT security is dedicated to **protecting physical systems** that **oversee and manage operations**.

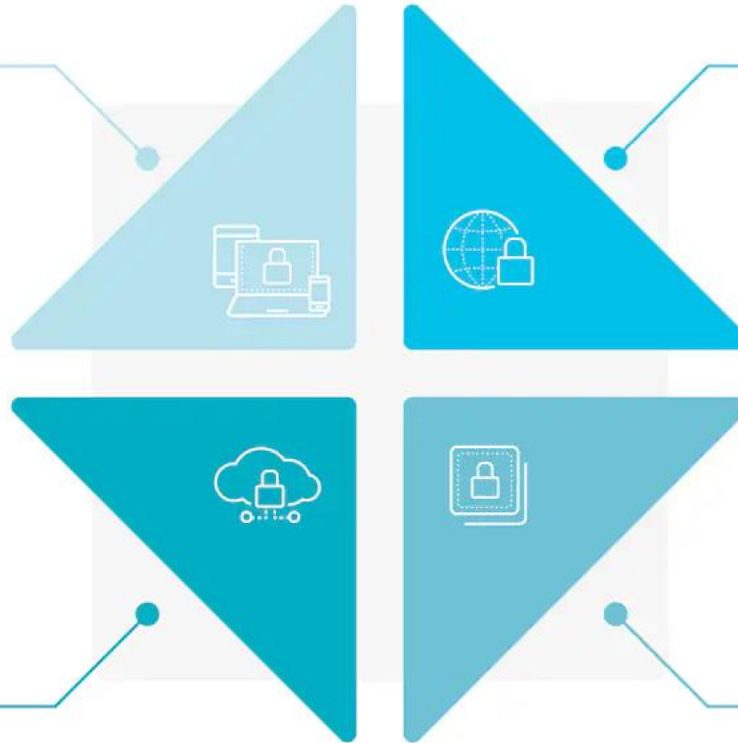
IT security

**Endpoint
security**

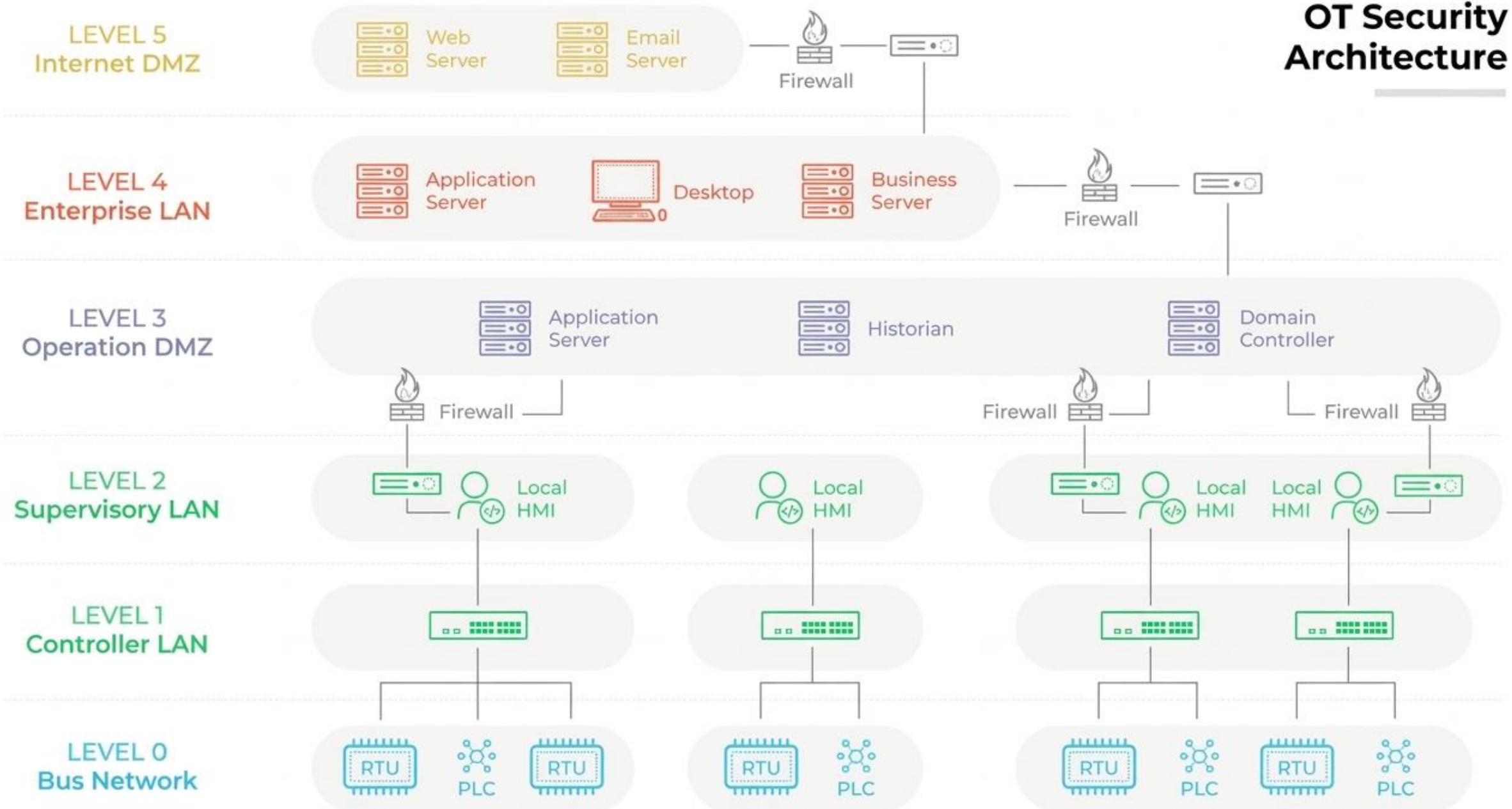
**Network
security**

**Cloud
security**

**Application
security**



OT Security Architecture



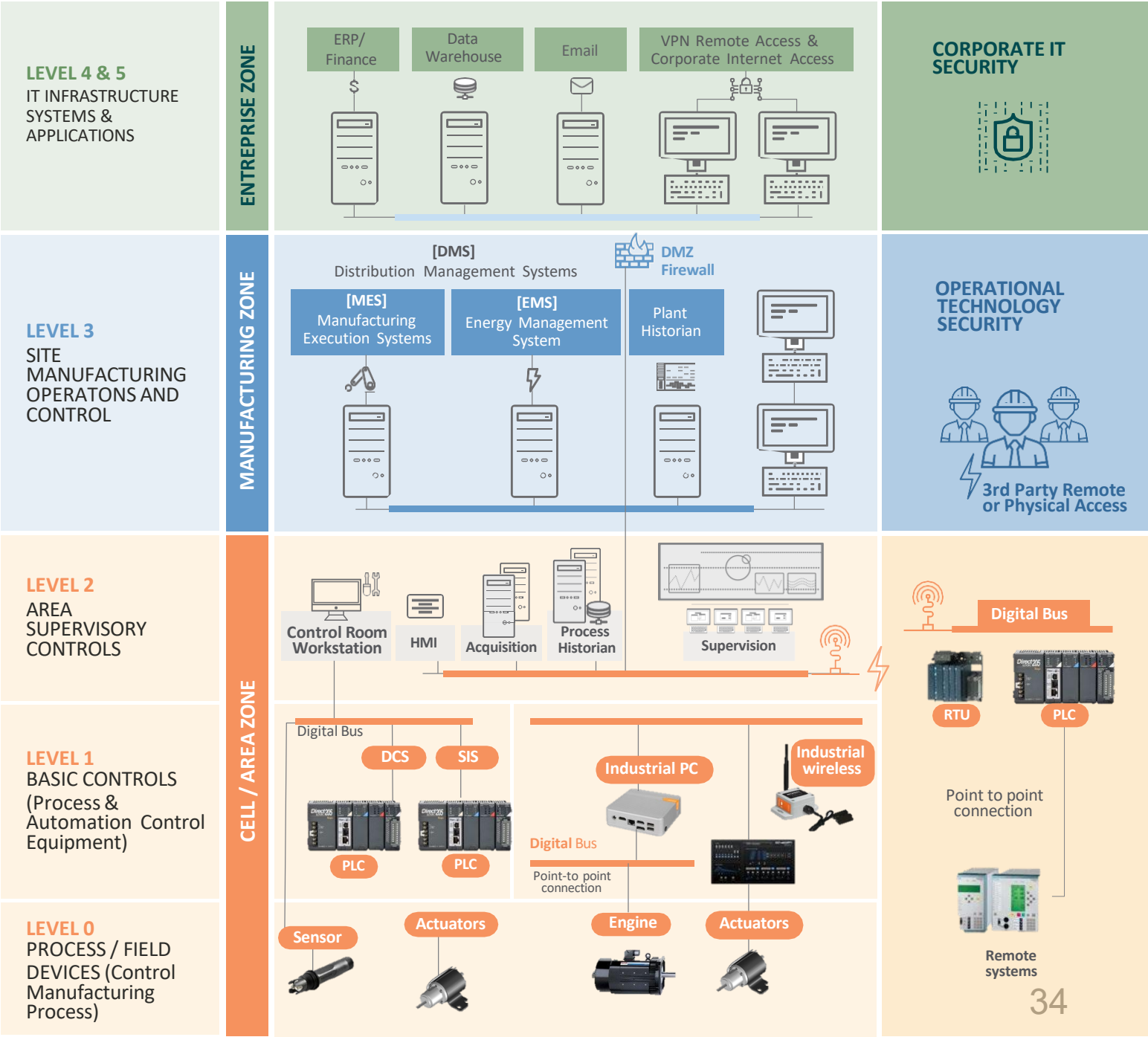
Purdue Logical Framework Model for IT / OT Security Convergence

LEVEL 3 - 1 CHALLENGES

- Legacy communications networks interfaces
- Unmanaged Ethernet switches/ lack of available ports
- Latency introduction due to SPAN
- Legacy unsupported OS
- ICS/OT system vendor certification requirements for changes
- Attacks from IT into OT
- 3rd Party Access to OT
- Limited asset Information accuracy

LEVEL 0 -1 CHALLENGES

- Legacy systems using proprietary log messages and event triggers
- Hard wired interfaces for signaling
- Serial messaging / signal-based OT protocols



OT Incidents vs IT Incidents

OT INCIDENTS

POWER OUTAGE

- Unauthorized control commands to IEDs cause multiple breaker trips, isolating substations, and cascading power failure.

PRODUCTION OUTAGE

- Malicious malware that changes PLC programming causes outage.

LOSS OF LIFE

- Safety control PLCs (emergency shutdown systems) take offline by malware downloaded to engineering workstation. Part of a coordinated malicious attack. Explosion occurs when threshold temperatures exceeded.

IT INCIDENTS

DENIAL OF SERVICE

- Infected computers on the internet request services from a website.
Website shuts down due to extreme load.

SENSITIVE DATA LOSS

- Phishing of CEO's laptop leads to theft of financial data, resulting in reputation loss, and stock price fall.

COMPANY IT ASSETS DELETED

- Malicious insider attains unauthorized access to Windows domain controller, and sends malware to corrupt/wipe all computing resources. Users, call center, partners, and clients loss all services.

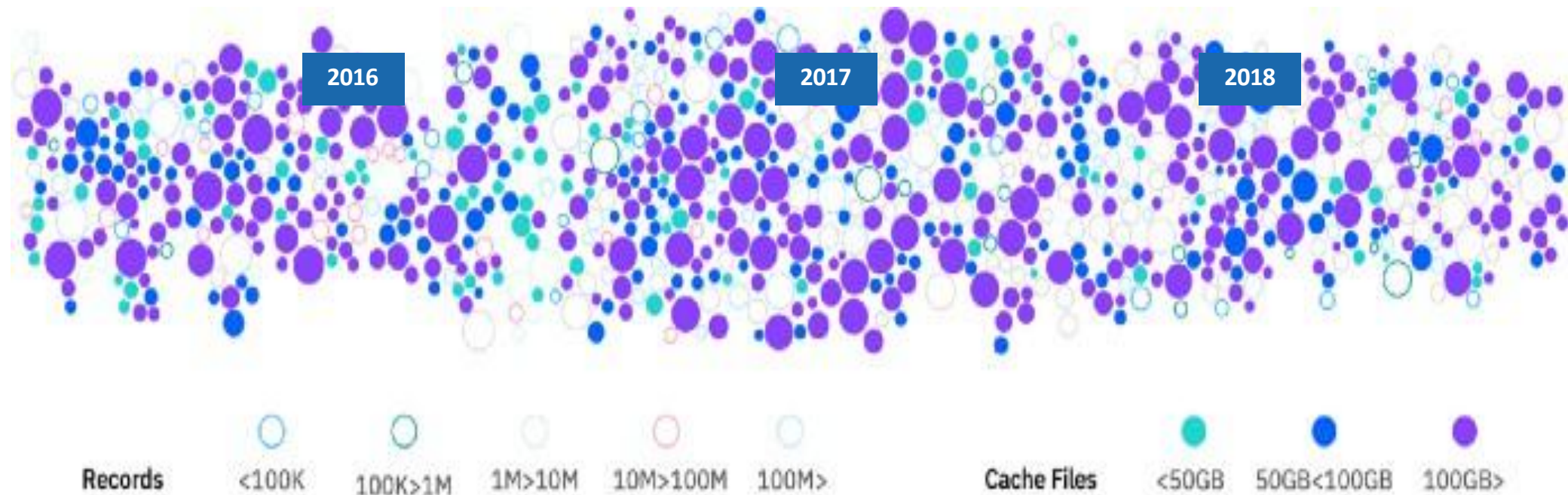
ATTACKER'S FOCUSING ON STEALING DATA

BILLIONS OF RECORDS AND HUNDREDS OF GIGABYTES

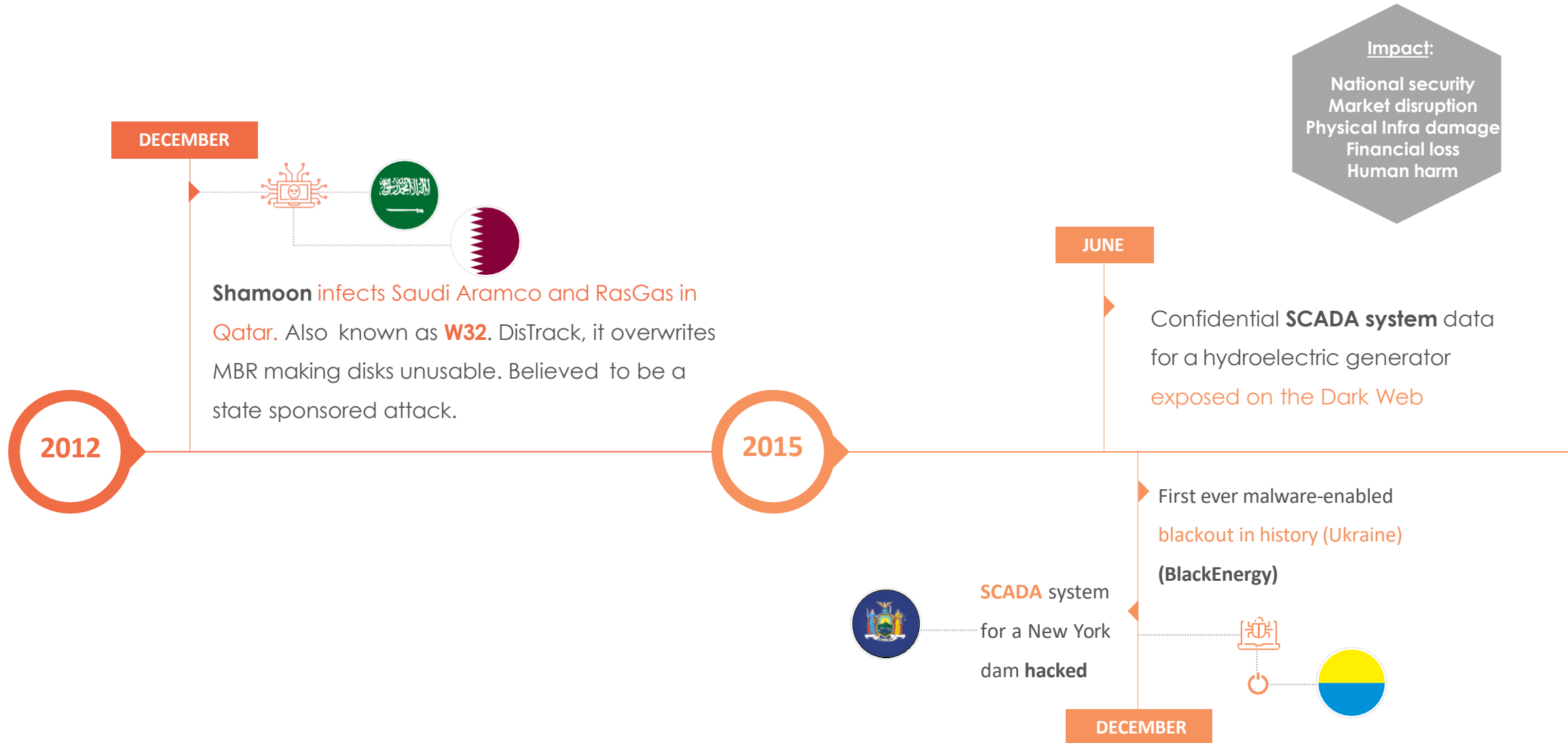
Over the last three years, more than 11.7 billion records and over 11 Terabytes of data were leaked or stolen in publicly disclosed incidents.

Figure: Sampling of the Impact of Security Incidents by records and Cache Files Compromised, Time and Impact, 2016 through 2018

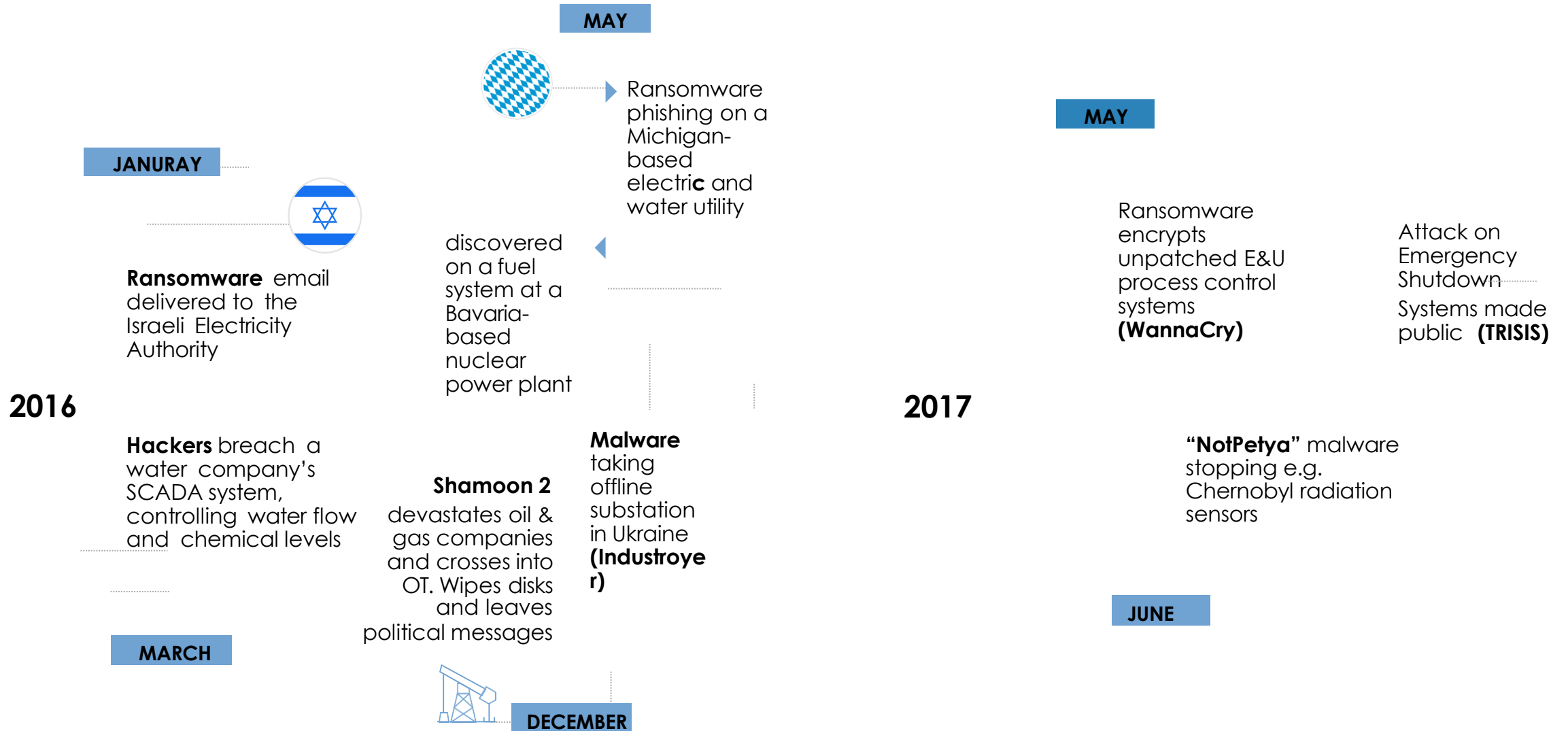
Source IBM X-Force



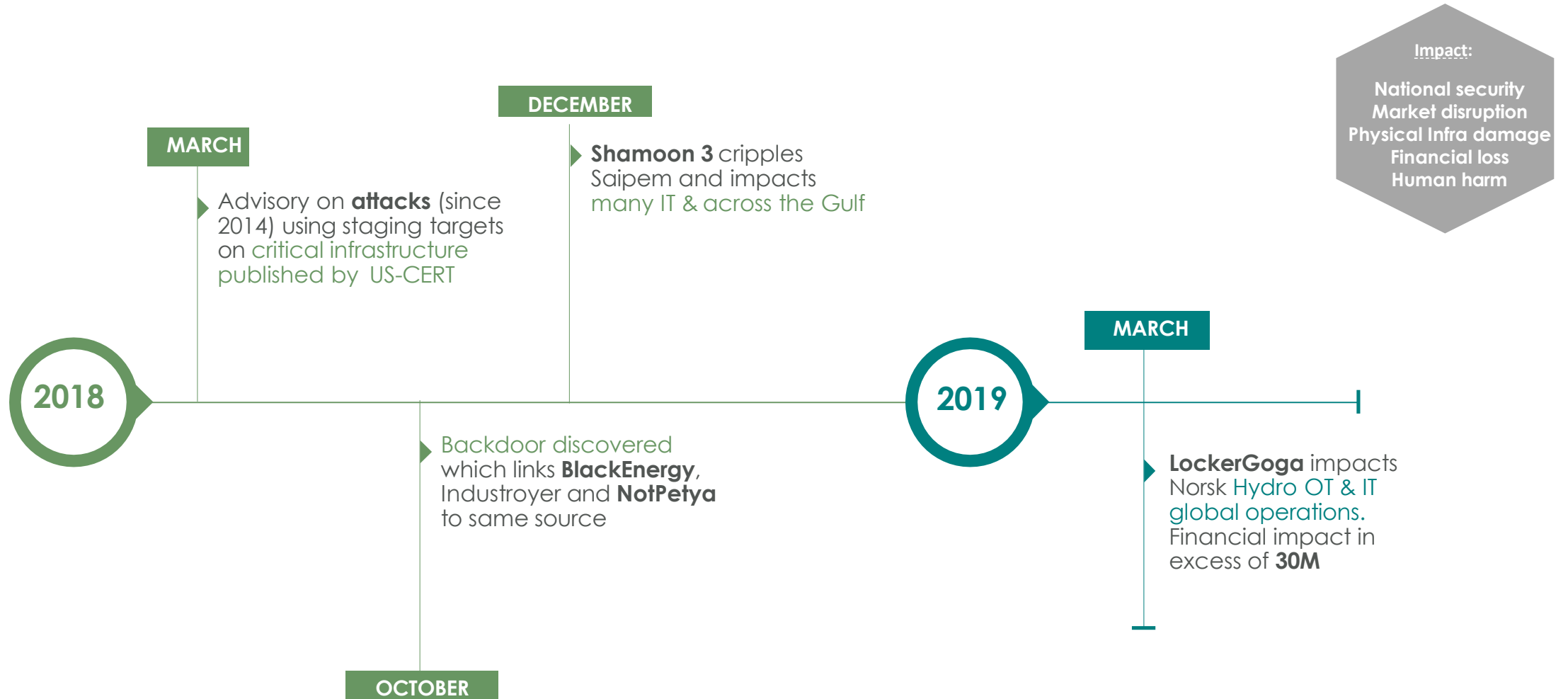
OT Threat landscape



OT Threat landscape



OT Threat landscape



MOST COMMON OT CYBERSECURITY MYTHS

MYTH 1

We don't connect to the Internet



MYTH 2

Control systems are behind a firewall



MYTH 3

Hackers don't understand control systems



MYTH 4

Our facility is not a target

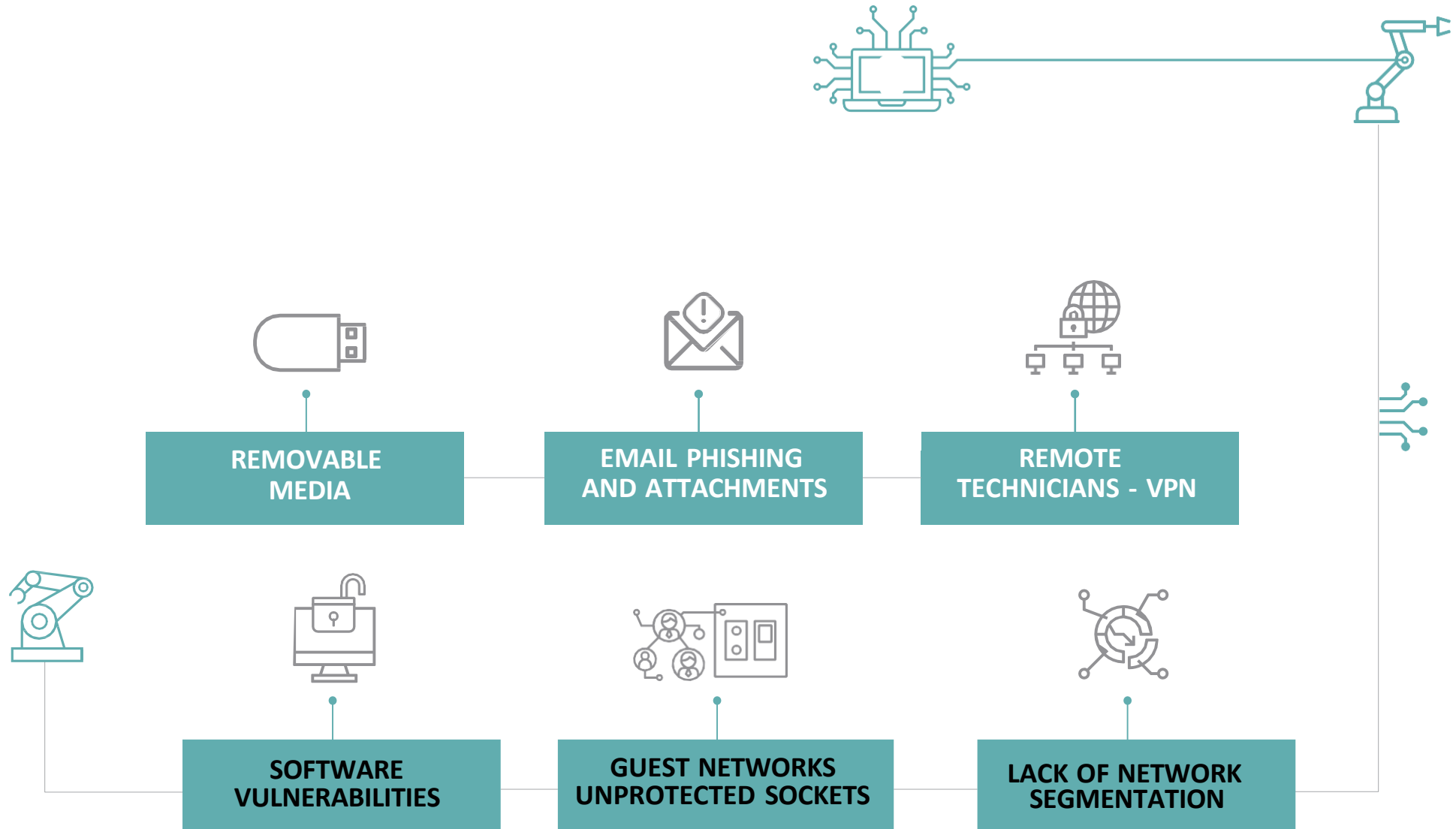


MYTH 5

Our safety systems will protect us



ATTACK VECTORS REACHING THE OT NETWORKS



SCADA functions

- Data acquisition
 - Analog and discrete values
- Event and alarm processing
 - Event and alarm
- Control
 - Tap changer
 - Shut capacitor/reactor
 - Switching devices
 - Generator excitation (AGC)
- Data storage, archiving and analysis

SCADA: Data Acquisition (Cont.)

- Measurements and Status Indications Collected are stored in a Real Time Database.
- The Values are Time tagged in the database.
- As new Values come in from the RTUs/IEDs old values are overwritten (or archived).

Monitoring and Event Processing

- **Events**

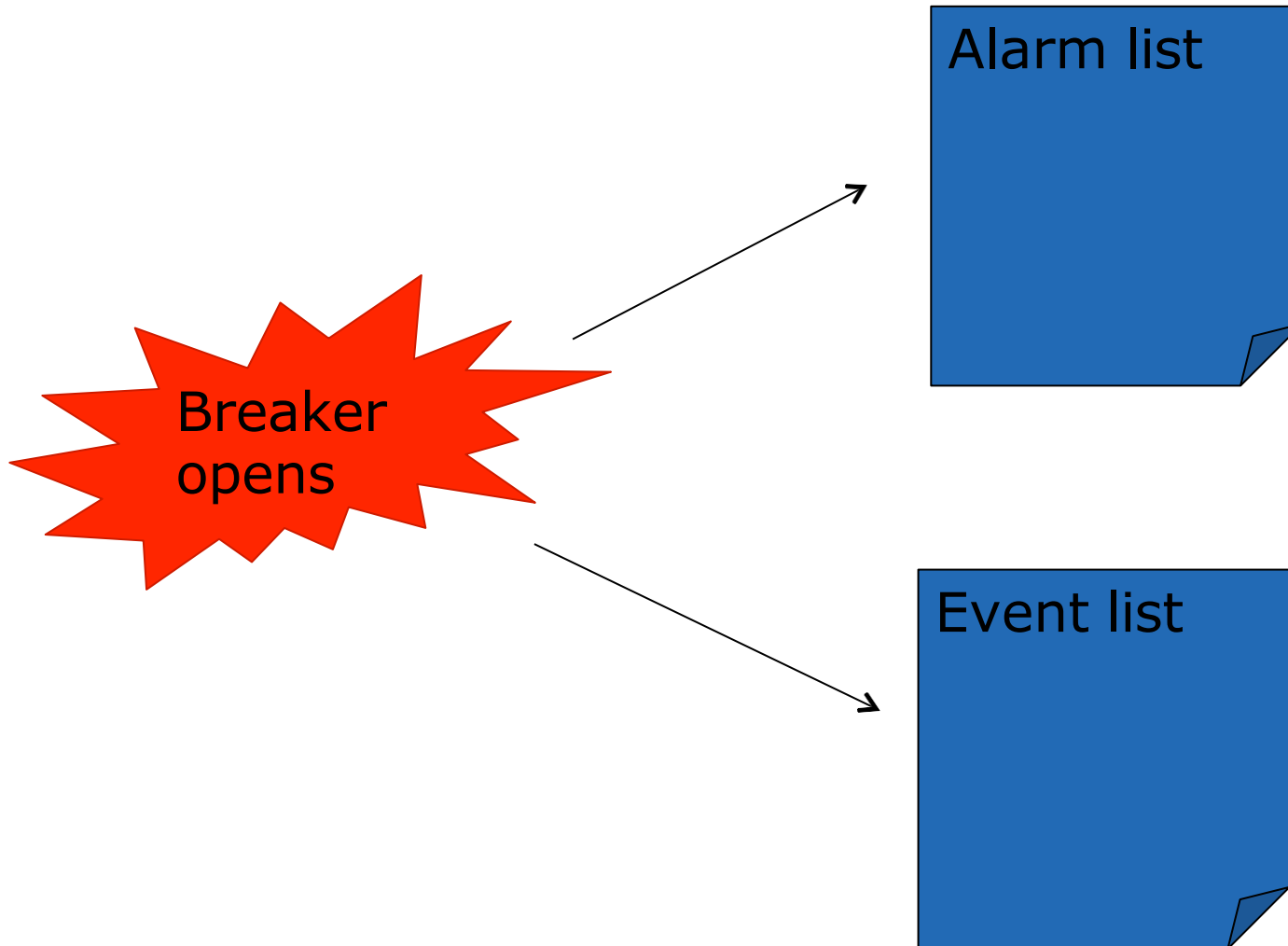
- Changing positions
 - Breaker / Disconnecter opens or closes
 - Value above/below a threshold
- Equipment activated
 - Reactor or capacitor engaged
- Automatic changes
 - On load tap changer changes state

- **Alarms**

- Critical events

- **It is a matter of definition**

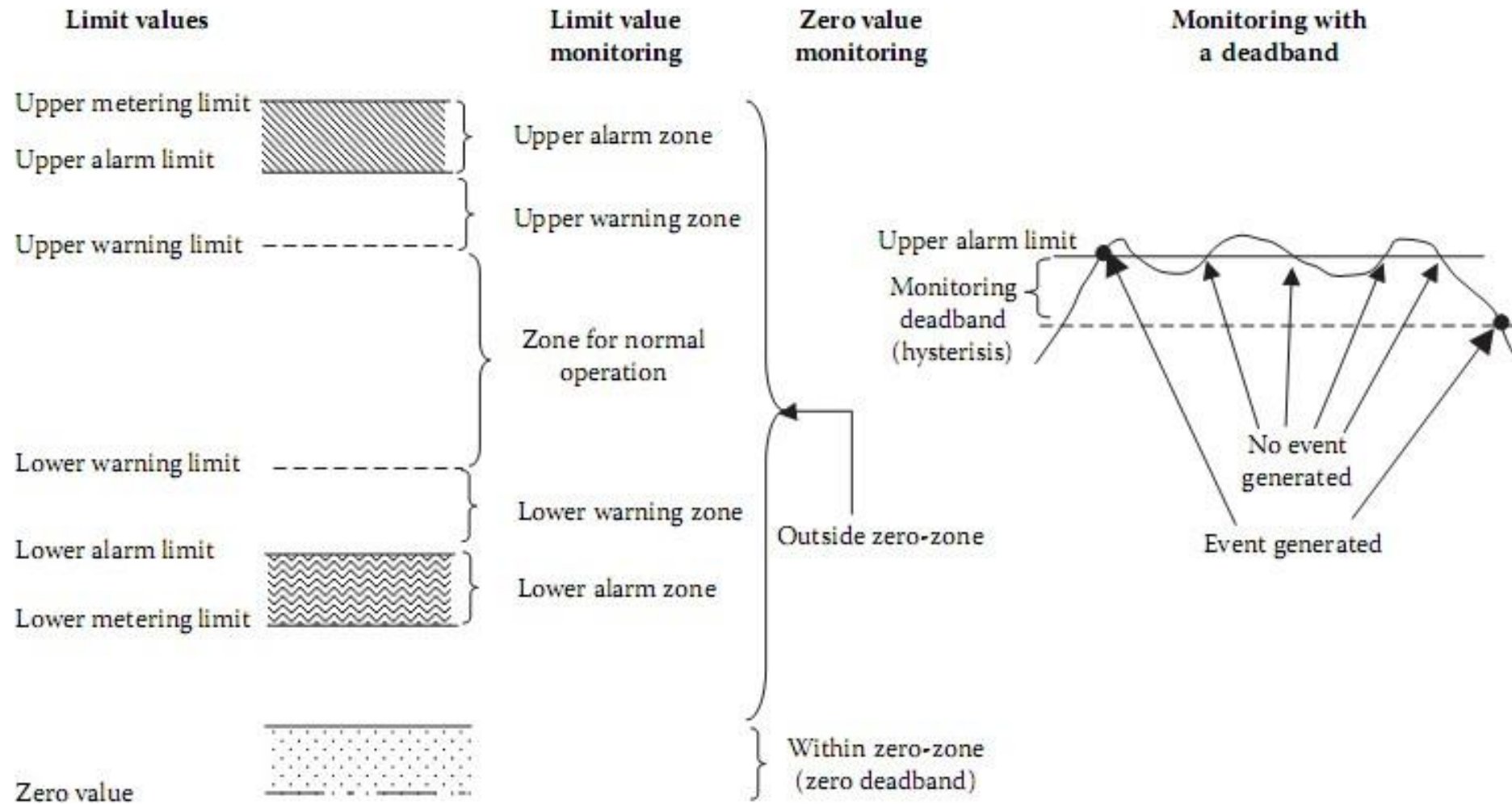
Alarm and Event Management



Alarm and Event Management

- Events
 - Changing positions
 - Breaker/Disconnecter opens or closes
 - Value above/below a threshold
 - Equipment activated
 - Reactor or capacitor engaged
 - Automatic changes
 - tap changer changes its position
- Alarms
 - Criticality
 - Sensitivity

Monitoring and Event Processing (cont.)



Time stamping

- Sequence of events is often important in analysis of chains of events
- **Time stamping of Events**
 - As close as possible to the source. For example the IED that collected the measurement
 - Requires time synchronisation of distributed devices
 - Additional Time-stamp at Front-End

SCADA: Data Storage, Archiving and Analysis

- **Data Collected from the process is sometimes archived, this due to many reasons:**
 - Regulations
 - Billing
 - Future Load planning
 - Performance Audits
 - Post Mortem Review, in case of disturbances or interruptions in the process.
 - Changed Values are “archived” at cyclic intervals, the interval depends on the importance of the values. Examples of cyclic intervals are: every scan interval, every 10 seconds or every hour.

Sequence of Events recorders

- Local function implemented in Substation Controller that keeps a record of all events in the substation
- Not all events are sent to the SCADA system
- SER logfiles can be uploaded to the SCADA system to enable analysis

SCADA: Control Functions

- **Individual Device Control**
 - Direct open/close commands to individual devices
 - **C**heck-**b**ack **b**efore **O**perate function.
- **Control Messages to Regulating Equipment**
 - E.g. Raising or lowering tap changer taps
- **Sequential Control**
 - E.g. in the case of a set of sequential switching steps to restore power through predefined backup configuration.
- **Automatic Control**
 - Triggered by an event or lapse of specific time period that invokes a control action
 - E.g. automatically changing load tap changer due to voltage set point violation

Non functional requirements

Functional requirements specifies what a system suppose to do and **Non functional requirements** specifies how a system should behave.

- **Availability:** the ratio of uptime to the sum of downtime and uptime.
- **Maintainability:** Repairing time for hardware and software
- **Scalability:** How easy the system could be expand
- **Security:** See SCADA security guest lecture
- **Interoperability/Openness:** how easy can the system be integrated with systems from other vendors

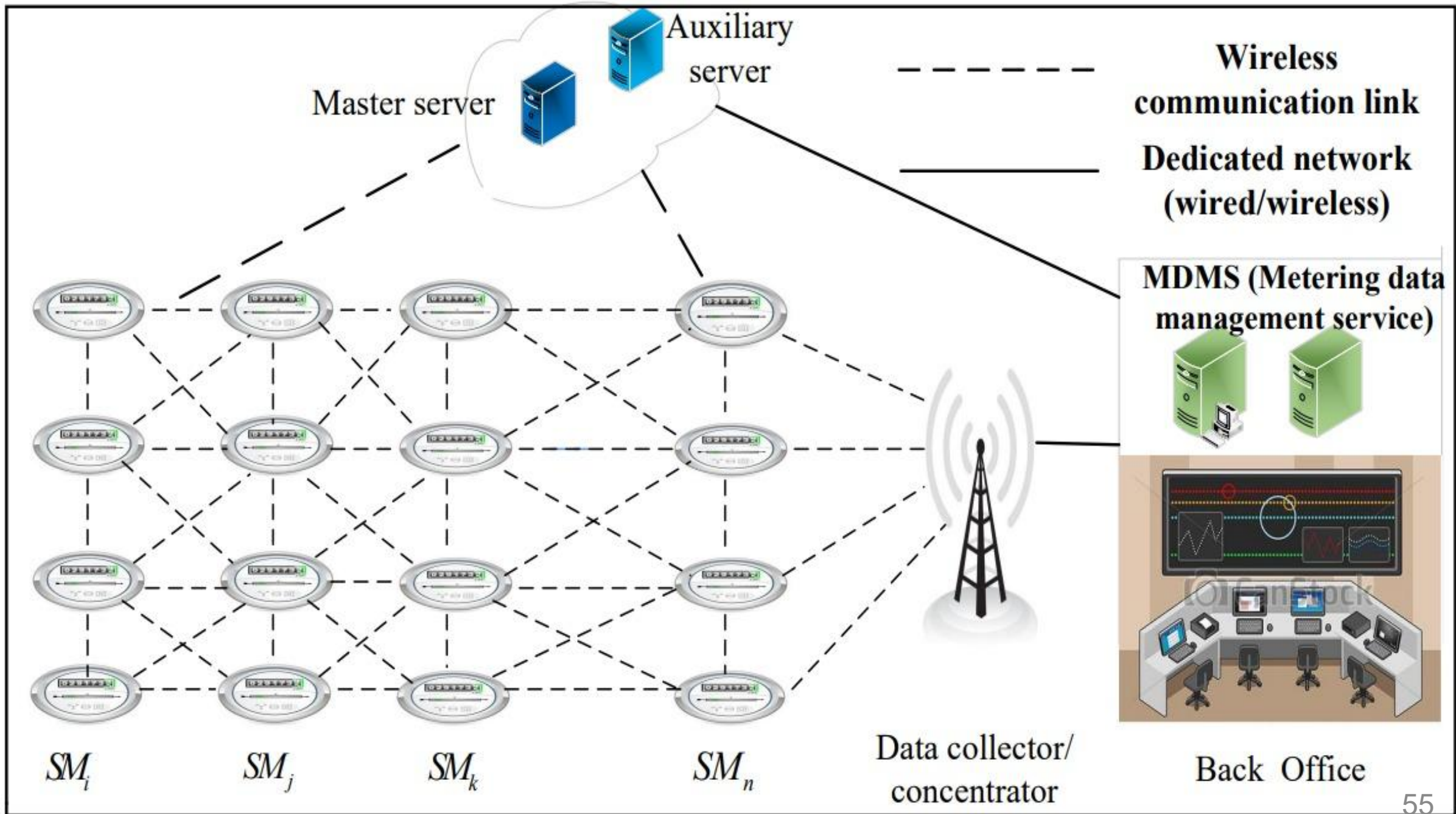
Status Update Performance Requirements

	Typical control services performance requirements						
		Execution time (s)	Accuracy %	Unavailability (h/mo)	Latency (s)	Single point /multiple point	Feedback sequence
Enterprise/function	Example measured elements						
Tier 1							
Substation operator control	Circuit breaker, capacitor switcher	2	99.99	4.0	1	Single	SBO
Auto-sectionalizing	Substation or field device	2	99.99	4.0	5	Multiple	None
Generation dispatch		2	99.9	4.0	1	Multiple	None
Substation algorithm		0.5	99.99	4.0	Depends	Depends	Depends

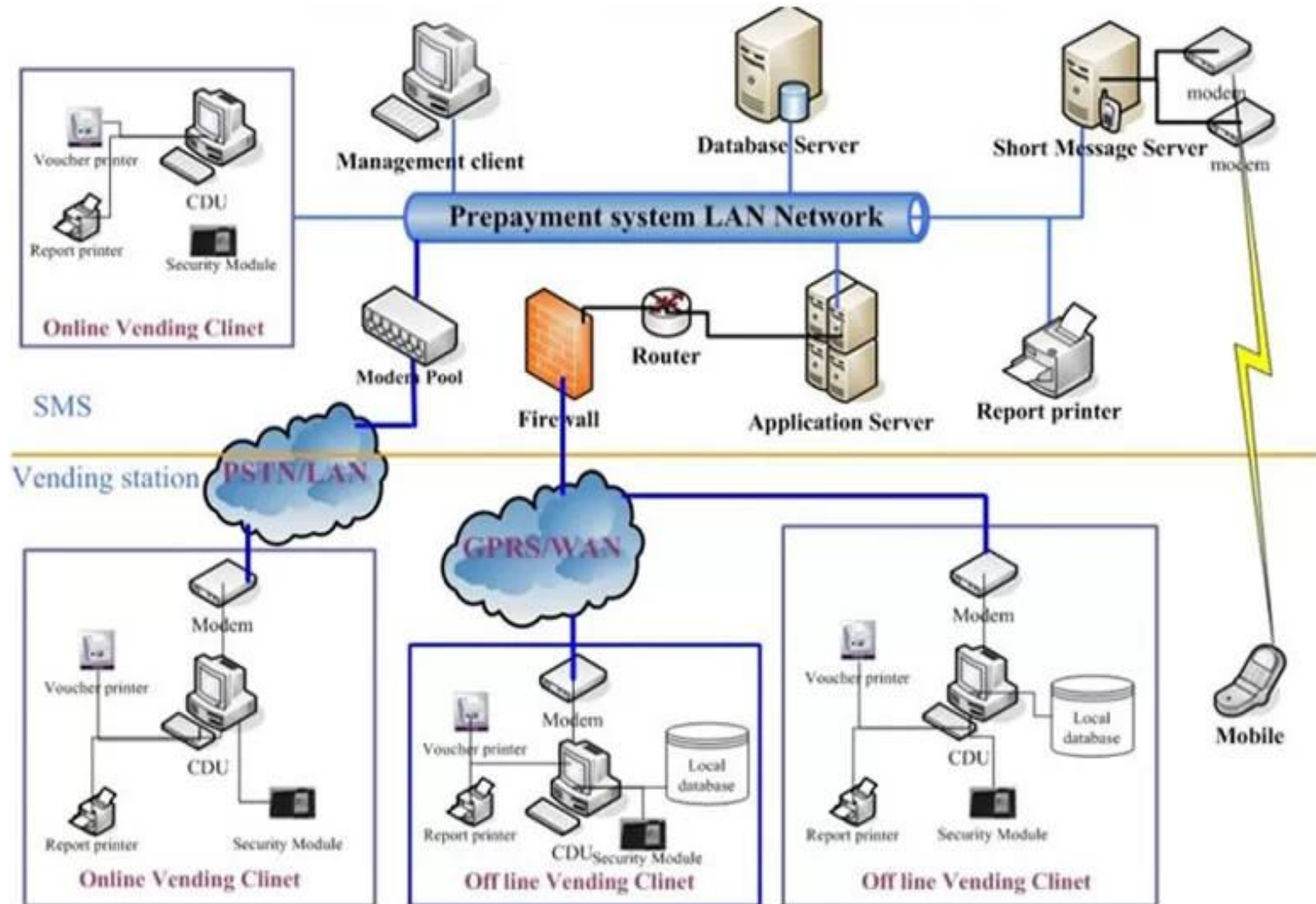
Reference: C37.1-2007 IEEE Standard for SCADA and Automation Systems

SMART METERING

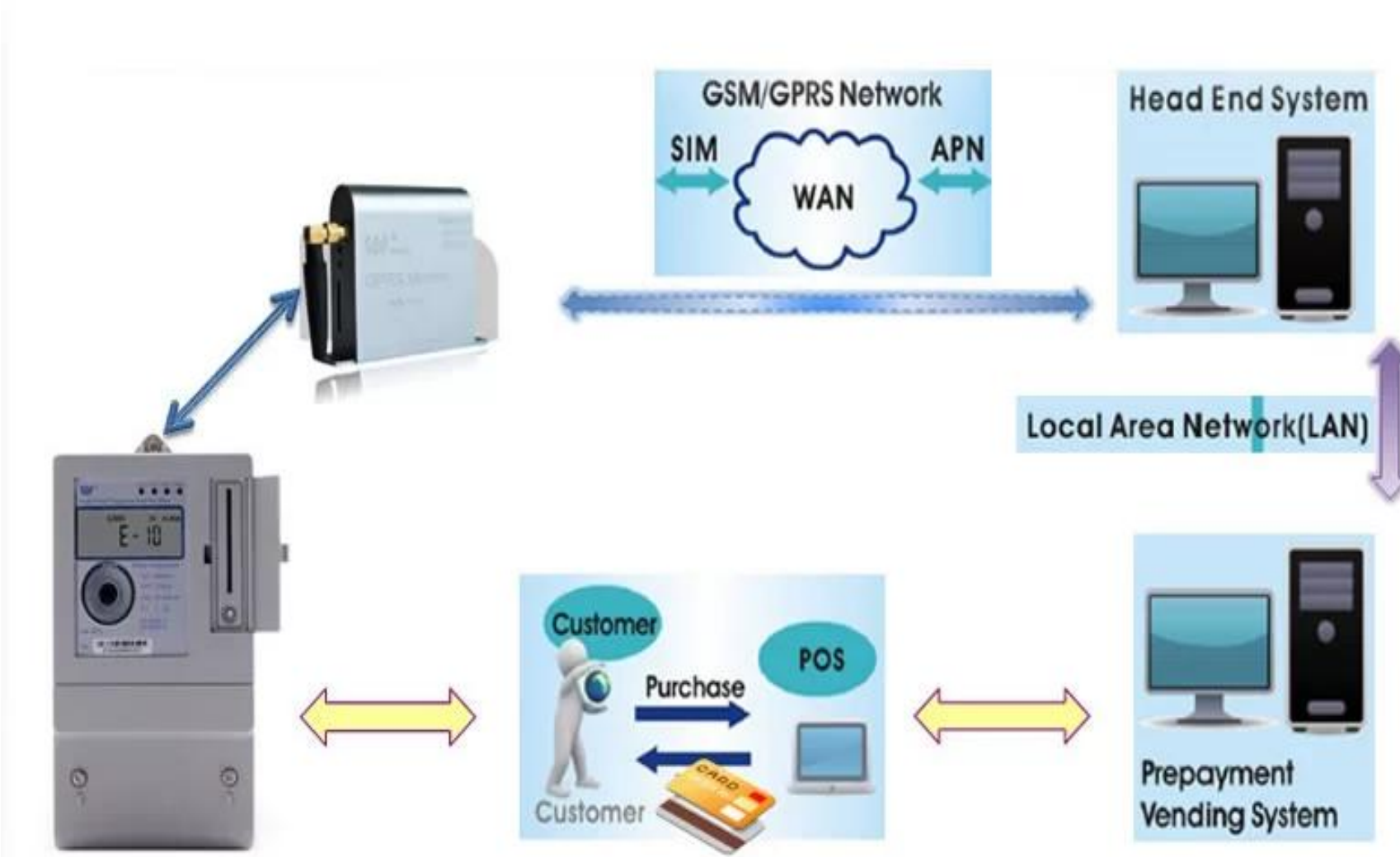
SMART METERING



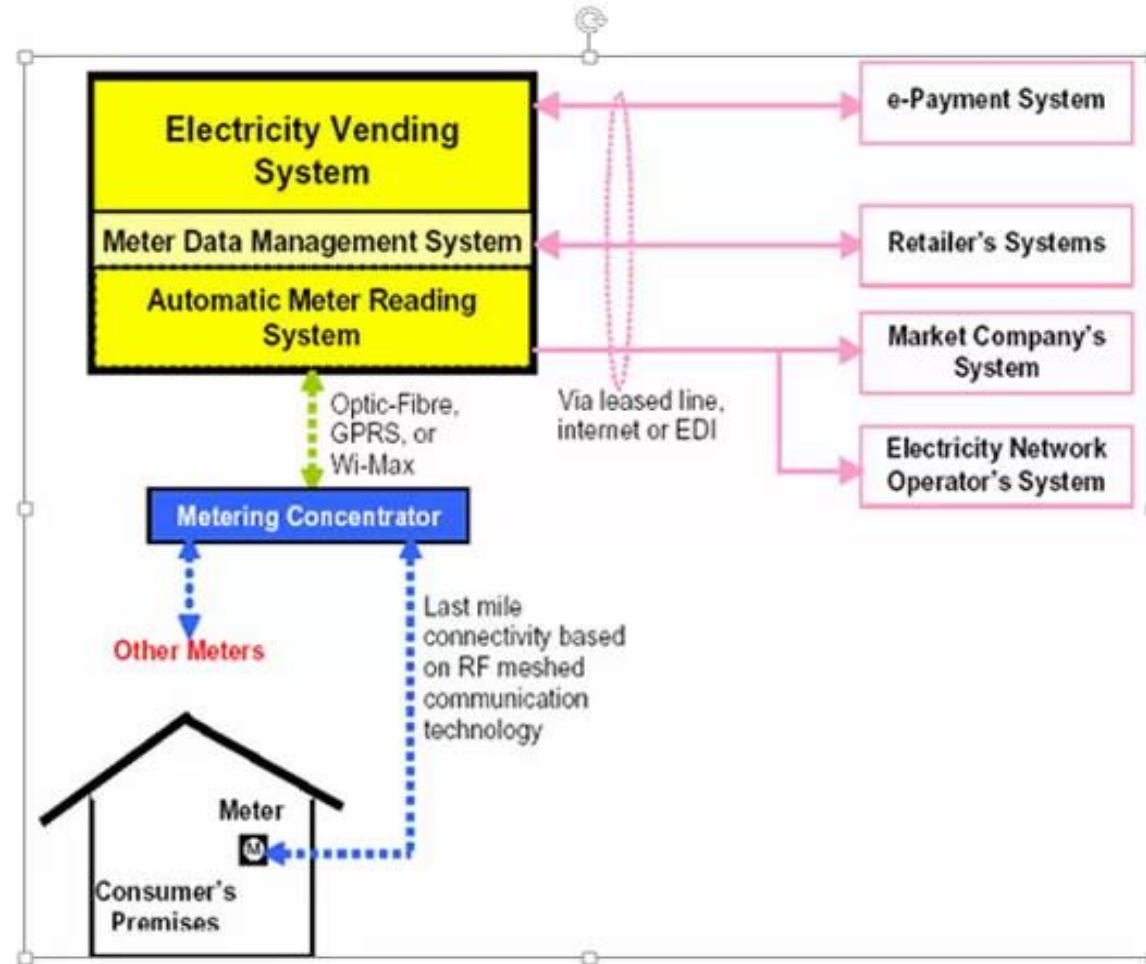
Smart Meter Network



Smart Meter with Vending System



Smart Meter with Vending System



SCADA ATTACKS

Attacks Targeting SCADA & Secure Power-Grid Protection

- Meter Encryption
- RTU/IED Hardening
- Secure Procurement
- Monitoring
- MFA

SCADA Attack Surface

- Field devices: smart meters, RTUs, PLCs, IEDs
- Control layer: HMI, engineering stations, historians
- Communications: serial, Ethernet, radio, cellular, IP
- Enterprise–OT connections increase exposure

Common Attacks Against SCADA

- Unauthorized access & credential theft
- Malware, ransomware & remote-access abuse
- Command injection / unauthorized control
- Man-in-the-middle & protocol manipulation
- Denial-of-service and communication disruption

Smart Meter & DLMS/COSEM Risks

- Meter data can reveal sensitive consumption patterns
- Weak authentication enables unauthorized access
- Unprotected traffic can expose or alter messages
- Compromised meters can become an entry point
- Legacy deployments may lack strong cryptography

Meter Encryption — DLMS/COSEM

- Use DLMS/COSEM security mechanisms appropriately
- Encrypt sensitive meter communications
- Authenticate meter, head-end and client
- Protect keys throughout their lifecycle
- Use secure key rotation and revocation

DLMS stands for **Device Language Message Specification**.

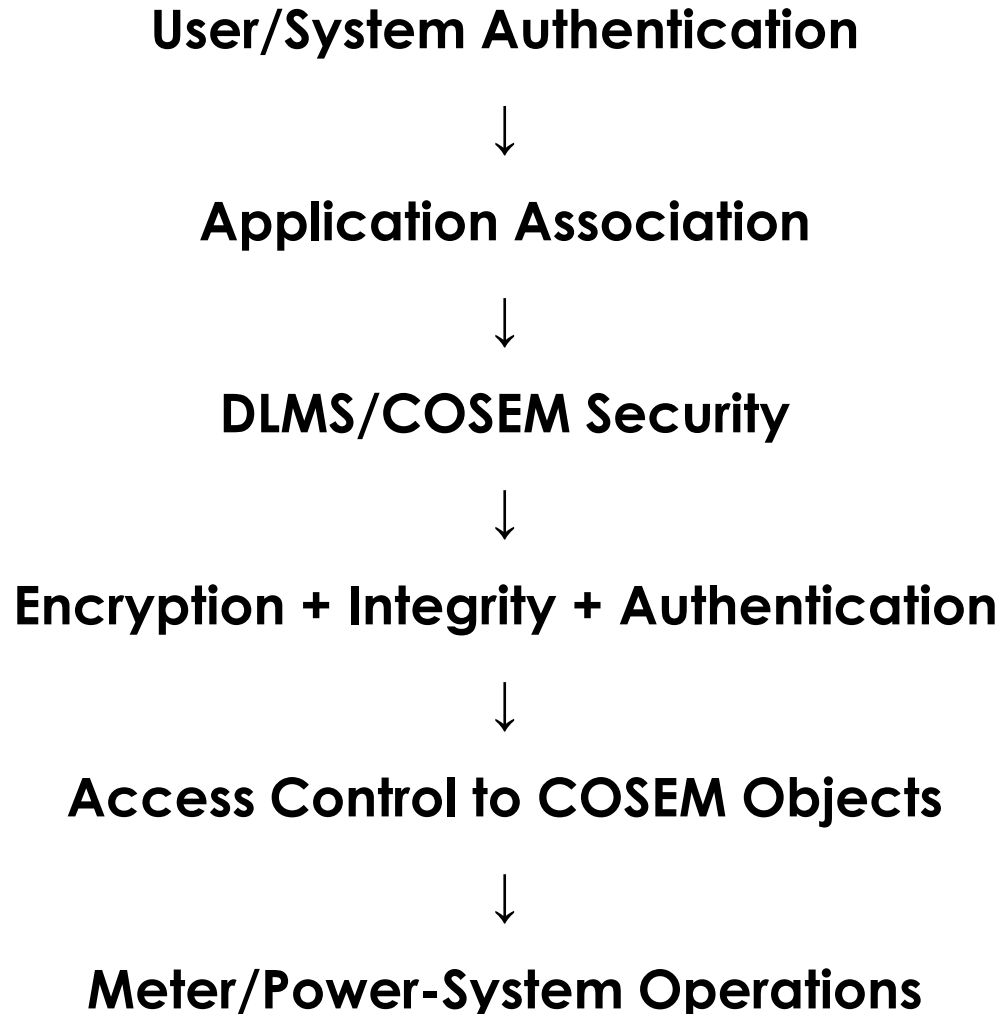
It is an international communication protocol/specification used primarily for **smart meters and utility devices** to exchange data securely and consistently.

- **DLMS** = Device Language Message Specification
- **COSEM** = Companion Specification for Energy Metering

Meter Encryption — DLMS/COSEM

Security aspect	Typical mechanism
Authentication	LLS / HLS
Confidentiality	AES encryption
Integrity	GMAC / authentication tag
Key establishment	Cryptographic key agreement
Key transport/wrapping	AES-based mechanisms
Access control	COSEM access rights

Meter Encryption — DLMS/COSEM



DLMS/COSEM security is not simply “encryption.” It combines authentication, cryptographic protection, integrity checking, key management, and object-level access control to secure smart-meter communications.

DLMS/COSEM (IEC 62056)

RTU / IED Hardening

- Disable unused ports, services and protocols
- Change default credentials immediately
- Apply signed firmware and controlled updates
- Restrict engineering access to authorized users
- Use allowlisting and secure configuration baselines

Secure SCADA Network Architecture

- Segment IT, OT and critical control zones
- Use firewalls between security zones
- Minimize inbound connections to OT
- Control remote access through monitored gateways
- Apply least privilege and deny-by-default rules

Secure Procurement

- Specify cybersecurity requirements before purchase
- Require secure-by-design products and protocols
- Demand vulnerability disclosure & patch support
- Require unique credentials and cryptographic support
- Define security testing and acceptance criteria

Identity & MFA

- Use MFA for remote and privileged access
- Separate operator, engineer and administrator accounts
- Use strong, unique credentials
- Apply role-based access control
- Review privileged access regularly

Monitoring & Detection

- Continuously monitor critical OT communications
- Baseline normal commands and traffic
- Detect unusual login, configuration and protocol activity
- Centralize security logs and alerts
- Synchronize time for reliable investigation

Incident Mitigation

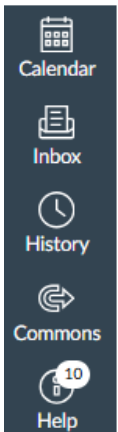
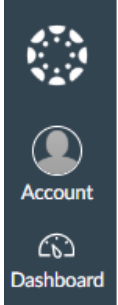
- Isolate affected systems without unsafe shutdowns
- Block compromised accounts and access paths
- Preserve logs and forensic evidence
- Restore from trusted configurations and backups
- Validate safety and control functions before recovery

Defense-in-Depth Checklist

- Encrypt + authenticate communications
- Harden RTUs, IEDs, meters and servers
- Segment networks and restrict remote access
- MFA + least privilege for users
- Monitor continuously and test incident response

References

- <https://rahmanziaur.github.io/pdb/>
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<https://canvaslite.instructure.com/enroll/WT49FB>
- <https://plcsimulationsoftware.com/>
- <https://www.paloaltonetworks.com/cyberpedia/ot-vs-it-security>
- <https://www.dragos.com/blog/ukraine-power-grid-cyberattack-crashoverride-10-year-lessons>
- https://github.com/mkhorasani/arduino_python_scada_system
- <https://industrialcyber.co/expert/why-scada-and-dcs-face-different-cyber-threats/>
- <https://repository.mut.ac.ke/items/663c10e0-e4b1-4b5c-b2f8-f200c9b247ca>
- <https://arxiv.org/pdf/1709.08505>
- <https://www.paloaltonetworks.com/cyberpedia/ot-vs-ics-vs-scada-security>
- <https://zpesystems.com/operational-technology-security-zs/>



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- Announcements
- Assignments
- Discussions
- Marks
- People
- Pages
- Files
- Syllabus
- Outcomes
- Rubrics
- Quizzes
- Modules
- BigBlueButton
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Module 1 — Foundations, Threat Landscape & Case Studies

Prerequisites: Introduction

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Module 2 — Prevention: Secure by Design

Prerequisites: Introduction, Module 1 — Foundations, Threat Landscape & Case Studies

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Module 3 — Detection, Mitigation & Recovery

Prerequisites: Introduction, Module 1 — Foundations, Threat Landscape & Case Studies, Module 2 — Prevention: Secure by Design

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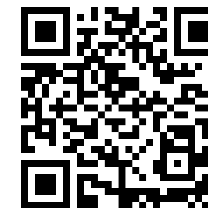
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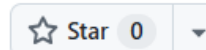
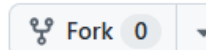


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CANVAS COURSE

- <https://rahmanziaur.github.io/pdb/>
- <https://github.com/rahmanziaur/pdb>
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Course GitHub Repo

⚡ Cybersecurity for Power Grid & SCADA/OT Systems

Professional Certificate Course for Assistant Engineers — Kaptai Power Grid, Bangladesh

Duration 30 Hours
 Level Professional Certification
 Modules 3
 Quizzes 3 × 10 Marks
 Curriculum Outcome Based (OBE)

Bloom's Taxonomy Revised (2001)

About

As a part of the training program offered by the BPDB Kaptai.

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Contributors 1

CLO–PO mapping

H = High, M = Medium, L = Low contribution

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CLO1	H	M			L		M
CLO2	M	H		L	M		M
CLO3	M	M	H	M	M		
CLO4	H		H	H			
CLO5	M	H	H		M	M	
CLO6	M	H		H		L	
CLO7	M	M	H		H	H	
CLO8	M	M	H	M	H	H	M

Outcome → delivery → assessment alignment

CLO	Teaching method	Assessment evidence
CLO1	Lecture, architecture walkthrough, Lab 1.1	Quiz 1 (items 1–5), Lab 1.1 worksheet
CLO2	Case-study discussion, ATT&CK-for-ICS mapping, Lab 1.2	Quiz 1 (items 6–10), ATT&CK mapping sheet
CLO3	Demonstration, hands-on Lab 2.1	Quiz 2 (items 1–4), Lab 2.1 RBAC matrix
CLO4	Simulator-based Lab 2.2	Quiz 2 (items 5–7), Lab 2.2 report
CLO5	Audit checklist, Lab 2.3, Lab 2.4	Quiz 2 (items 8–10), audit and procurement worksheets
CLO6	Pcap/log analysis, Lab 3.1	Quiz 3 (items 1–4), Lab 3.1 findings
CLO7	Workflow design, Labs 3.2 and 3.3	Quiz 3 (items 5–8), workflow and SOP deliverables
CLO8	Tabletop exercise, Lab 3.4	Quiz 3 (items 9–10), IR plan and exercise debrief

Module 1 — Foundations, Threat Landscape & Case Studies (10 h)

Focus: Understand what we are protecting, who and what threatens it, and what real attacks teach us.

Course GitHub Page

- <https://rahmanziaur.github.io/pdb/>
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*Thank
You*